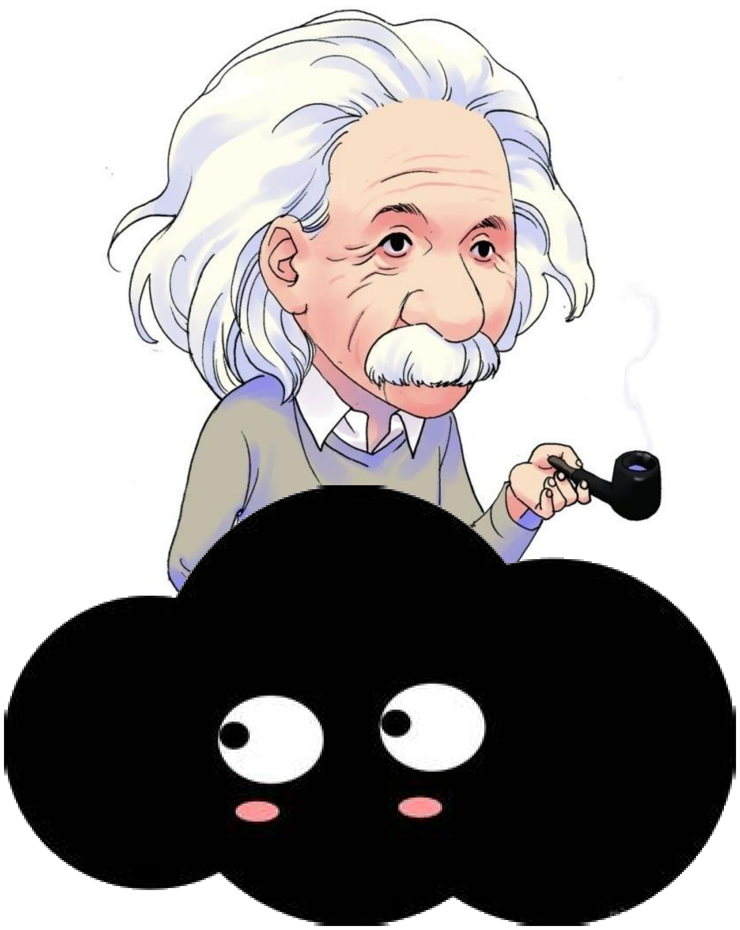
From two clouds to two storms to two revolutions



[Theory of relativity]

[Quantum mechanics]

Planck's concept

↓

Einstein's Photoelectric Effect

↓

Bohr's model

….

↓

Many other great physics scientists

↓

Quantum Mechanics

Quantum Mechanics

Scientists	Year	Age	Discovery
Planck	1900	42	Planck postulate
Einstein	1905	26	Photoelectric effect
Bohr	1913	28	Hydrogen atom
de Broglie	1923	31	Matter wave
Pauli	1924	24	Spin
Heisenberg	1925	24	Matrix mechanics
Schrodinger	1926	37	Schrodinger equation
Born	1926	44	Born rule
Dirac	1928	26	Dirac equation

CHAPTER 38

Photons and Matter Waves

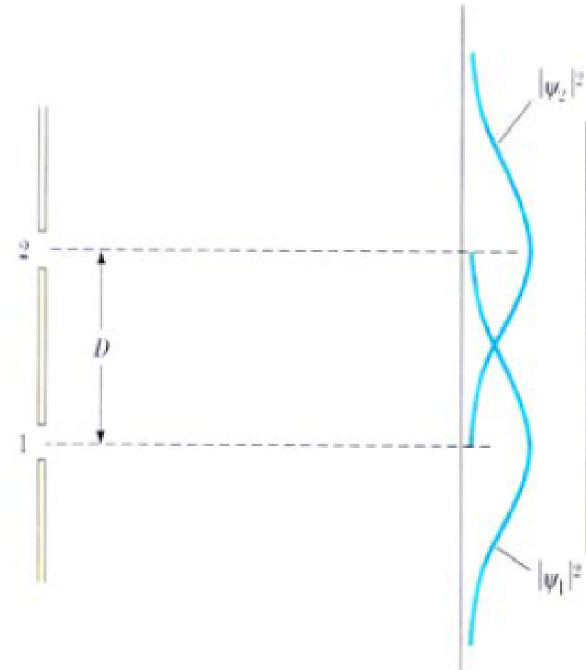
Chapter 38

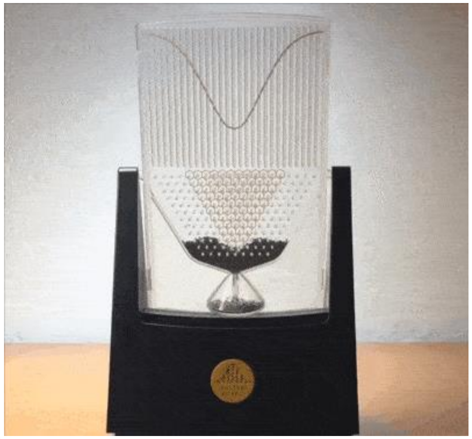
- <u>**Electron’s Double-Slit Experiment**</u>
- **Photons**
- **Matter Wave**
- **Tunneling Through a Potential Barrier**

- If one slit is covered during experiment, the result is a symmetric curve that peaks around the center of the open slit, much like the pattern formed by bullets shot through a hole in armor plate.
- These curves are expressed as:

$$|\psi_1|^2 = \psi_1^{*}\psi_1$$

$$|\psi_2|^2 = \psi_2^{*}\psi_2$$





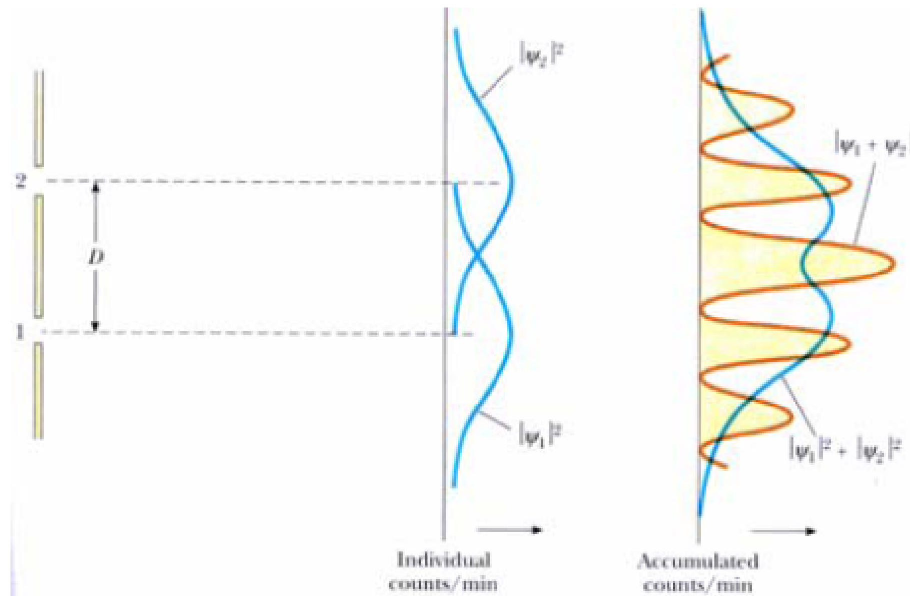
- The accumulated result is simply the sum of the individual results, and the interference pattern lost. This curve can be expressed as: $|\psi_1|^2 + |\psi_2|^2$

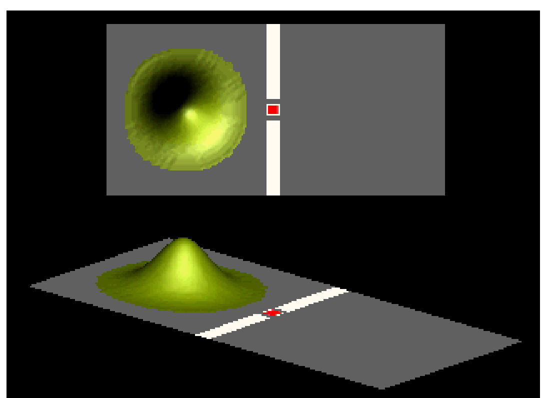
- The electron is in superposition state when both slits are open:

$$\psi = \psi_1 + \psi_2$$

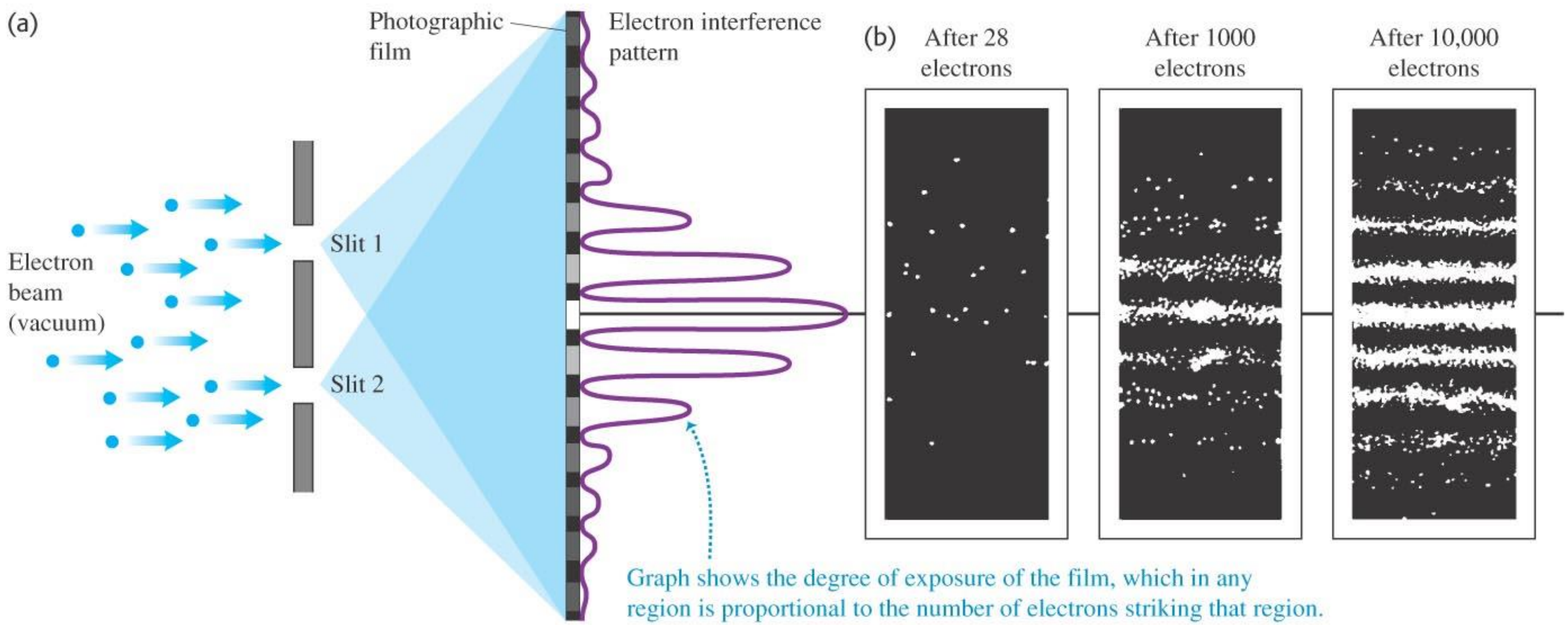
- The probability of detecting the electron at the detector is:

$$\psi^2 = |\psi_1 + \psi_2|^2$$

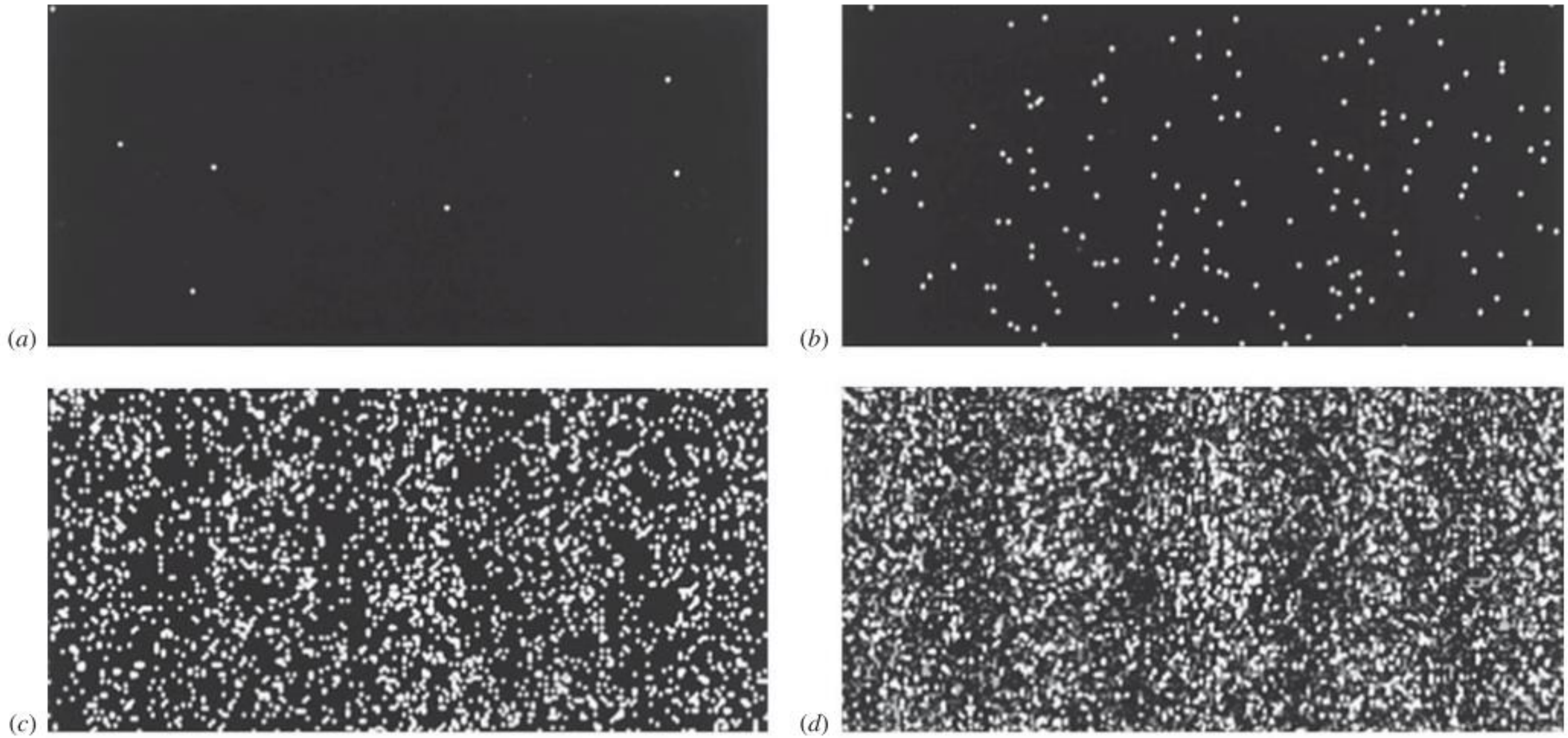


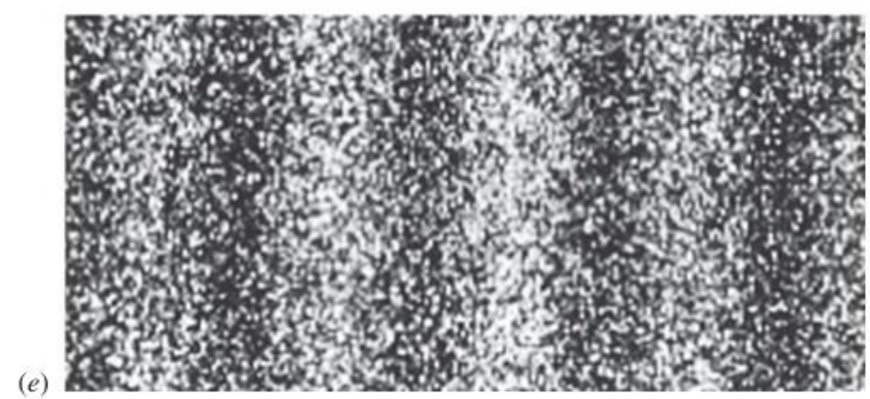


Double slits with electrons



Electrons are sent out one by one (showing wave nature)





Central Research Laboratory, Hitachi, Ltd., Kokubinju, Tokyo;
H. Ezawa, Department of Physics, Gakushuin University,
Mejiro, Tokyo

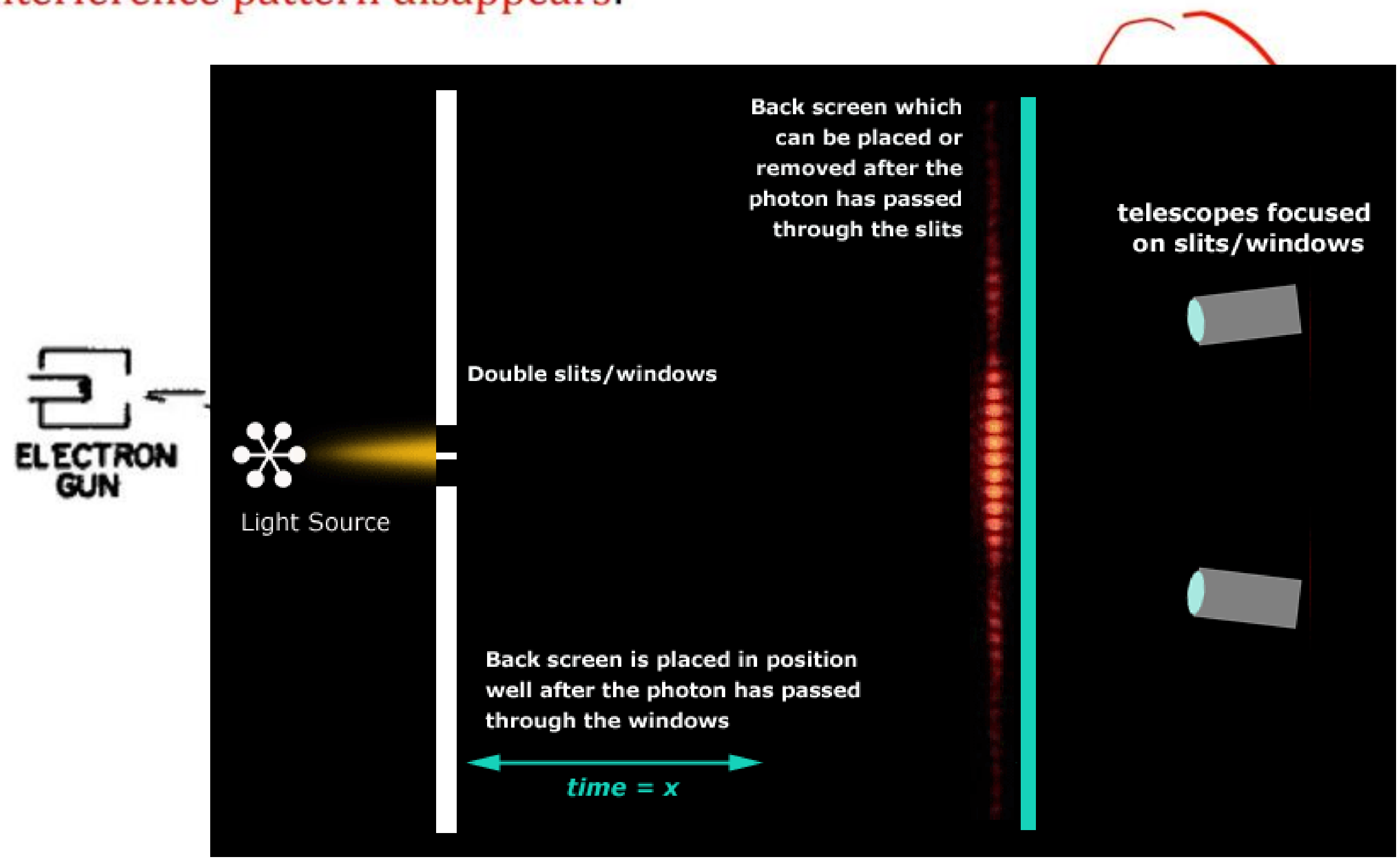
Figure 38-9 Photographs showing the buildup of an interference pattern by a beam of electrons in a two-slit interference experiment like that of Fig. 38-6. Matter waves, like light waves, are *probability waves.* The approximate numbers of electrons involved are (*a*) 7, (*b*) 100, (*c*) 3000, (*d*) 20 000, and (*e*) 70 000.

Double-slit Experiment with two Detectors

Let's try to determine which slit the electrons pass through by placing two cameras (detectors) near both slits.

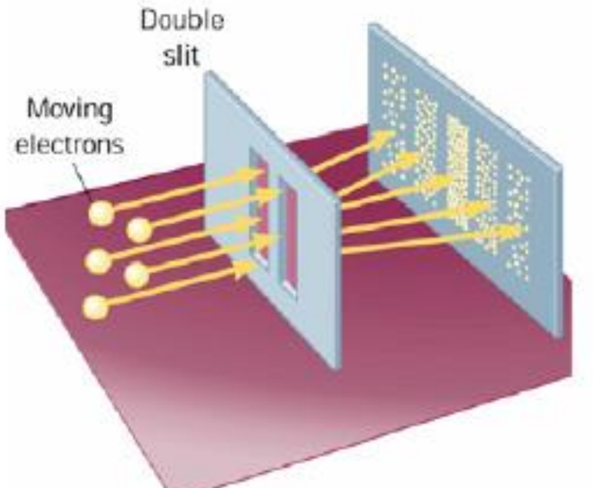
Yes, every time electron only passes one slit. The two detectors never click together at the same time.

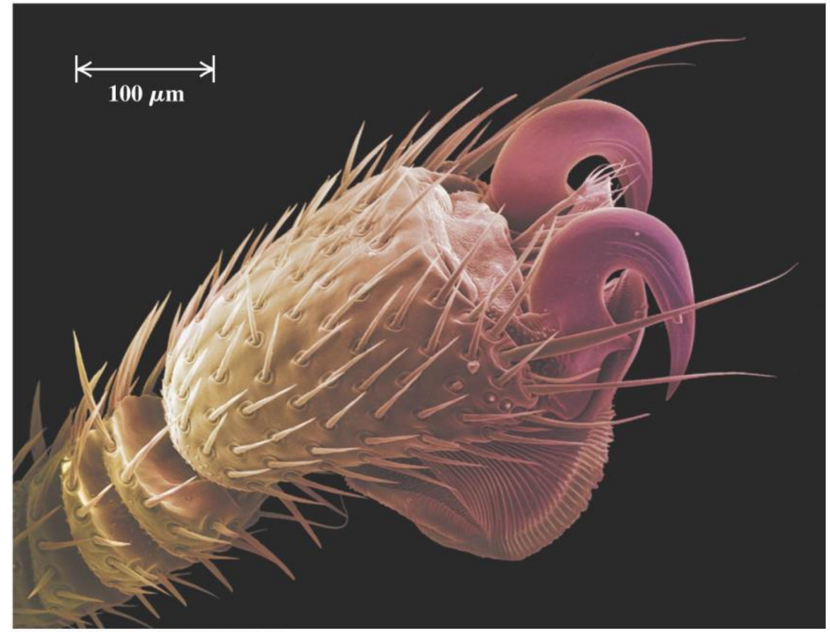
But… Interference pattern disappears!

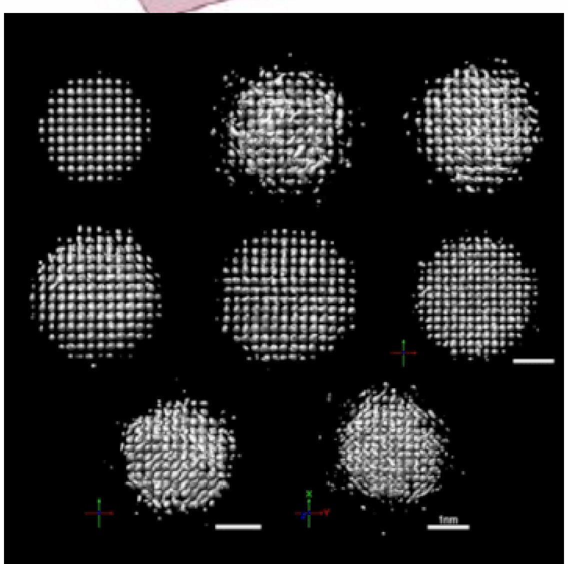


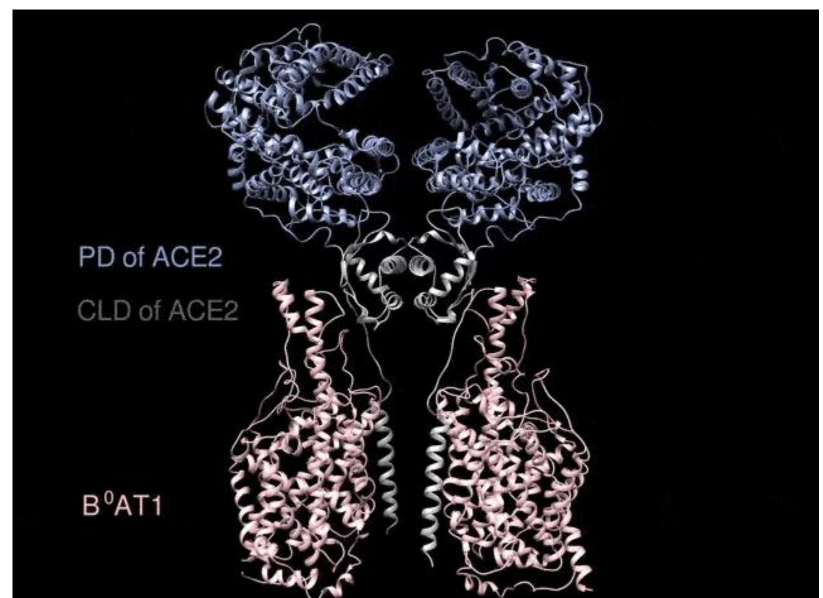
– Interference and diffraction patterns have been observed for helium atoms, hydrogen atoms and neutrons.

– The **electron microscope** is an important example of the dual nature of electrons.





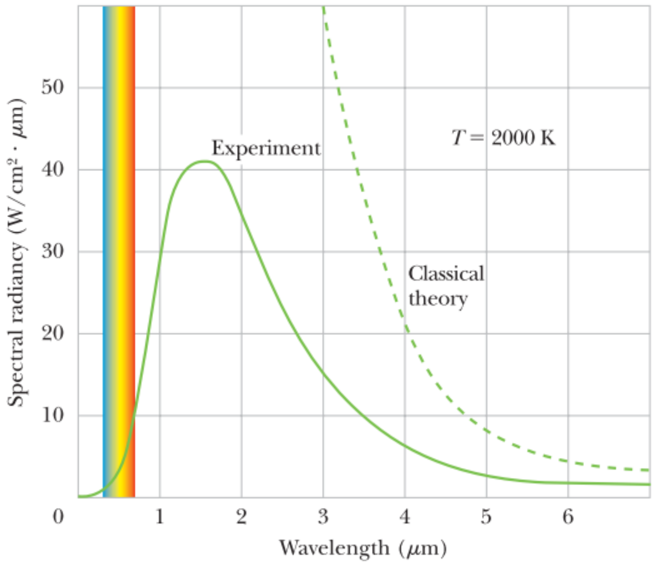


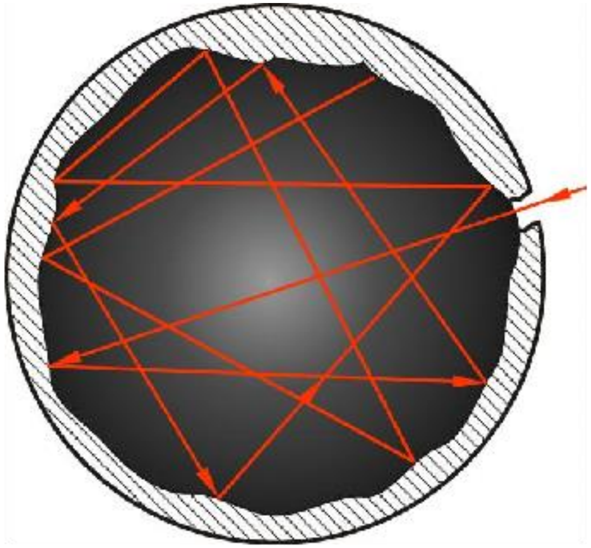


Chapter 38

- **Electron's Double-Slit Experiment**
- **Photons**
- **Matter Wave**
- **Tunneling Through a Potential Barrier**

From Blackbody radiation to Photoelectric effect





Experimental Setup. We can make an ideal radiator by forming a cavity within a body and keeping the cavity walls at a uniform temperature. The atoms on the inner wall of the body oscillate (they have thermal energy), which causes them to emit electromagnetic waves, the thermal radiation. To sample that internal radiation, we drill a small hole through the wall so that some of the radiation can escape to be measured (but not enough to alter the radiation inside the cavity). We are interested in how the intensity of the radiation depends on wavelength.

From Blackbody radiation to Photoelectric effect

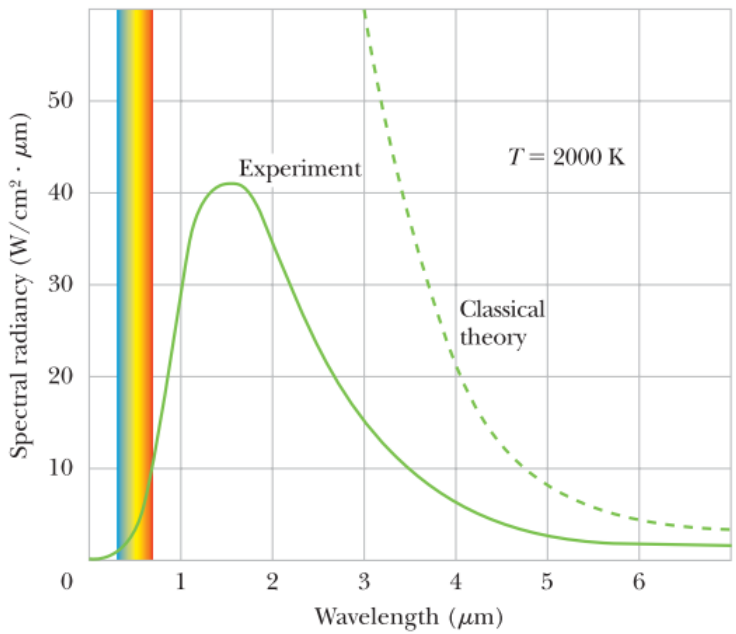
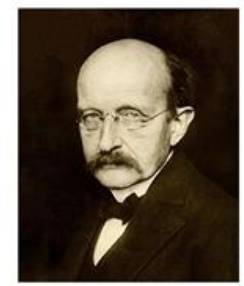


Figure 38-8 The solid curve shows the experimental spectral radiancy for a cavity at 2000 K. Note the failure of the classical theory, which is shown as a dashed curve. The range of visible wavelengths is indicated.

Classical radiation law:

$$S(\lambda) = \frac{2\pi ckT}{\lambda^4}$$

Max Planck (1858 –1947) German physicist; Nobel Prize in Physics in 1918.



Planck's Solution. In 1900, Planck devised a formula for $S(\lambda)$ that neatly fitted the experimental results for all wavelengths and for all temperatures:

Highlighted by Man-Hong Y

$$S(\lambda) = \frac{2\pi c^2 h}{\lambda^5} \frac{1}{e^{hc/\lambda kT} - 1} \quad \text{(Planck's radiation law).} \qquad (38\text{-}14)$$

From Blackbody radiation to Photoelectric effect

1. **普朗克的量子化假设：**

黑体以 $h\nu$为能量单位不连续地发射和吸收频率为ν的光子的能量.

且能量单位 $h\nu$称为能量子，h 为普朗克常量 ($h=6.62606896\times 10^{-34}J\cdot S$)

2. **普朗克公式的推导过程：**

2.1 **任意频率 ν 下的辐射能量：**

假设有一处于平衡状态的黑体，其内有数量为 N 的原子可吸收或发出频率为ν的光子，其中 N^g 为这些原子中处在基态的原子数，N^e 为处在激发态（此处指可由基态原子受频率为ν的光子激发达到的能态）的原子数，n 为频率为ν的光子平均数。则由统计力学中的麦克斯韦-玻尔兹曼公式[2]知：

$N_e \propto N e^{-\frac{E_e}{kT}}$　　$N_g \propto N e^{-\frac{E_g}{kT}}$ 由此可得

$$\boxed{\frac{N_e}{N_g}=e^{-\frac{Ee-Eg}{kT}}=e^{-\frac{h\nu}{kT}}} \qquad (2.1.1)$$

平衡状态下，体系内原子在两能级间相互转化的速率相等，且其速率正比于转化的概率和该状态下的原子数目。结合爱因斯坦系数关系[3]可得：

$$N_g n = N_e (n+1) \qquad (2.1.2)$$

结合(2.1.1)，可解得：

$$\boxed{n=\frac{1}{e^{\frac{h\nu}{kT}}-1}} \qquad (2.1.3)$$

则该状态下光子总能量为：

$$\varepsilon_0 = nh\nu = \frac{h\nu}{e^{\frac{h\nu}{kT}}-1} \qquad (2.1.4)$$

2.2 $\nu \sim \nu + d\nu$ **频率段中可被体系接收的频率数目**

设所求黑体为规整的立方体，其长，宽，高分别为L_x, L_y，L_z。体积为V_0。不妨先讨论一维情况：

体系线宽为 L，则 L 必为光子半波长的整数倍，设其波数为 K，有

$$k_j = \frac{j\pi}{L} \quad (j \text{ 为整数}) \qquad (2.2.1)$$

成立 。

则两相邻可被体系接收的频率所对应的波数间隔为

$$\delta k = k_{j+1} - k_j = \frac{\pi}{L}$$

(2.2.2)

由此可得在Δk 波数段内，可被体系接收的频率数目（或称波数数目）为：

$$\Delta N = \frac{\Delta k}{\delta k} = \frac{L}{\pi}\Delta k \qquad (2.2.3)$$

因空腔内光波为驻波（波数为 K 和-K 的两列波合成），考虑 K 值的正负 ，(2.2.3)式可修正为：

$$\Delta N = \frac{L}{2\pi}\Delta k \qquad (2.2.4)$$

由此可得，在三维情况下，有

$$\Delta N_x = \frac{L_x}{2\pi}\Delta k_x$$

$$\Delta N_y = \frac{L_y}{2\pi}\Delta k_y$$

(2.2.5)

From Blackbody radiation to Photoelectric effect

$$\Delta N_z = \frac{L_z}{2\pi}\Delta k_z$$

并由此得到

$$\boxed{\Delta N(k) = \Delta Nx \cdot \Delta Ny \cdot \Delta Nz = \frac{L_x L_y L_z}{8\pi^3}} \qquad \Delta k_x \Delta k_y \Delta k_z$$

(2.2.6)

因$L_x L_y L_z$为黑体体积V_0，$\Delta k_x \Delta k_y \Delta k_z$ 为 K 体积元d^3k，考虑半径为 K，厚度为dk的球壳，则(2.2.6)式可化为：

$$dN(k) = \frac{V_0}{8\pi^3}d^3k = \frac{V_0}{8\pi^3}4\pi k^2 dk$$

即$dN(k) = \frac{V_0}{2\pi^2}k^2 dk$ (2.2.7)

由 $k = \frac{2\pi v}{c}$ 代入(2.2.7)可得

$$dN(v) = \frac{4\pi v^3}{c^3}V_0\, dv \tag{2.2.8}$$

因光为电磁波，对任意波矢 K 可有两正交的偏振，其频率相互独立，所以(2.2.8)应修正为：

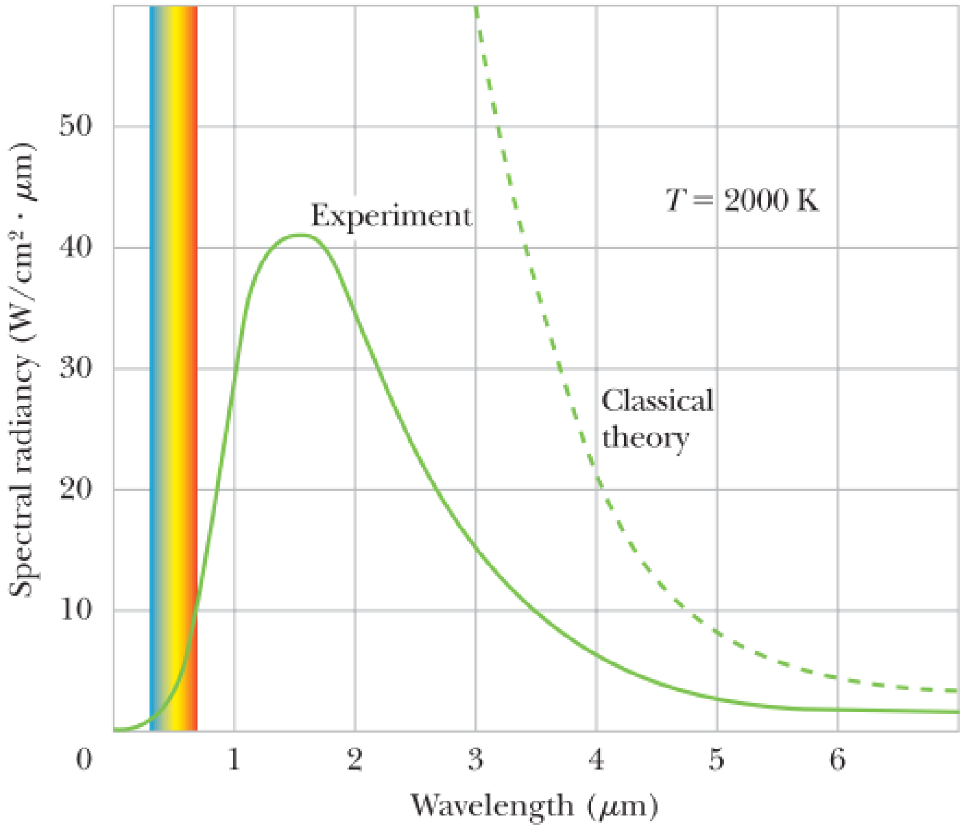
$$dN(v) = \frac{8\pi v^3}{c^3}V_0\, dv \tag{2.2.9}$$

此即为 $v \sim v + dv$ 频率段中可被体系接收的频率数目。

2.3 $v \sim v + dv$ 频率段内的黑体辐射能量

由(2.1.4)和(2.2.9)可得 $v \sim v + dv$ 频率段内的黑体辐射能量为：

$$\varepsilon_0 dN(v) = \frac{8\pi h v^3}{c^3(e^{\frac{hv}{kT}} - 1)}V_0\, dv$$



Planck

Planck's solution. In 1900, Planck devised a formula for $S(\lambda)$ that neatly fitted the experimental results for all wavelengths and for all temperatures.

$$\frac{8\pi h\nu^3}{c^3}\cdot\frac{1}{e^{\frac{h\nu}{kT}}-1}$$

$$S(\lambda) = \frac{2\pi c^2 h}{\lambda^5}\frac{1}{e^{hc/\lambda kT}-1} \quad \text{(Planck's radiation law).} \qquad (38\text{-}14)$$

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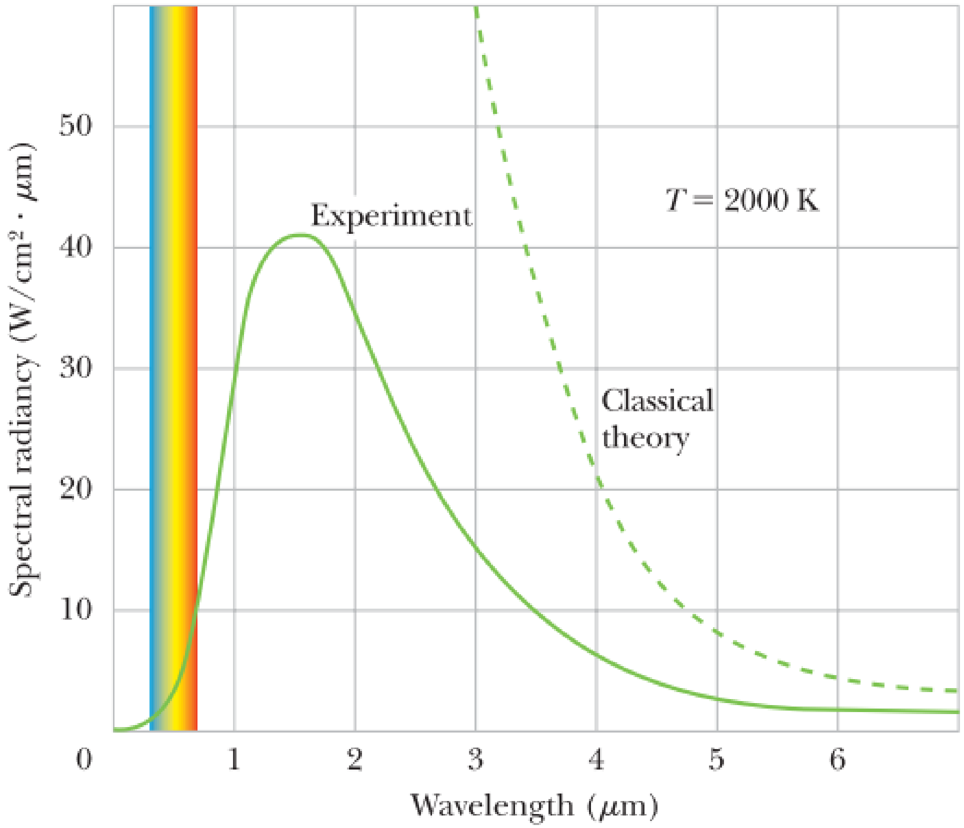
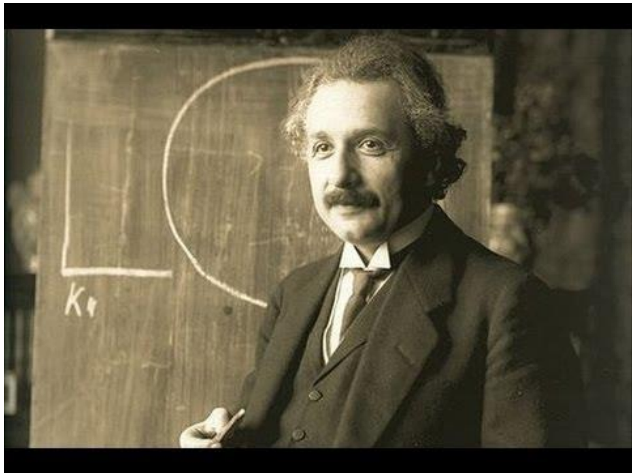




Einstein

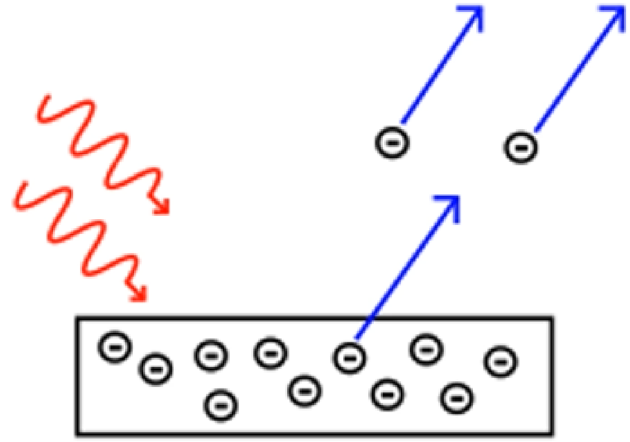
Einstein's Solution. No one understood Eq. 38-14 for 17 years, but then Einstein explained it with a very simple model with two key ideas: (1) The energies of the cavity-wall atoms that are emitting the radiation are indeed quantized. (2) The energies of the radiation in the cavity are also quantized in the form of quanta (what we now call photons), each with energy $E = hf$. In his model he explained the processes by which atoms can emit and absorb photons and how the atoms can be in equilibrium with the emitted and absorbed light.

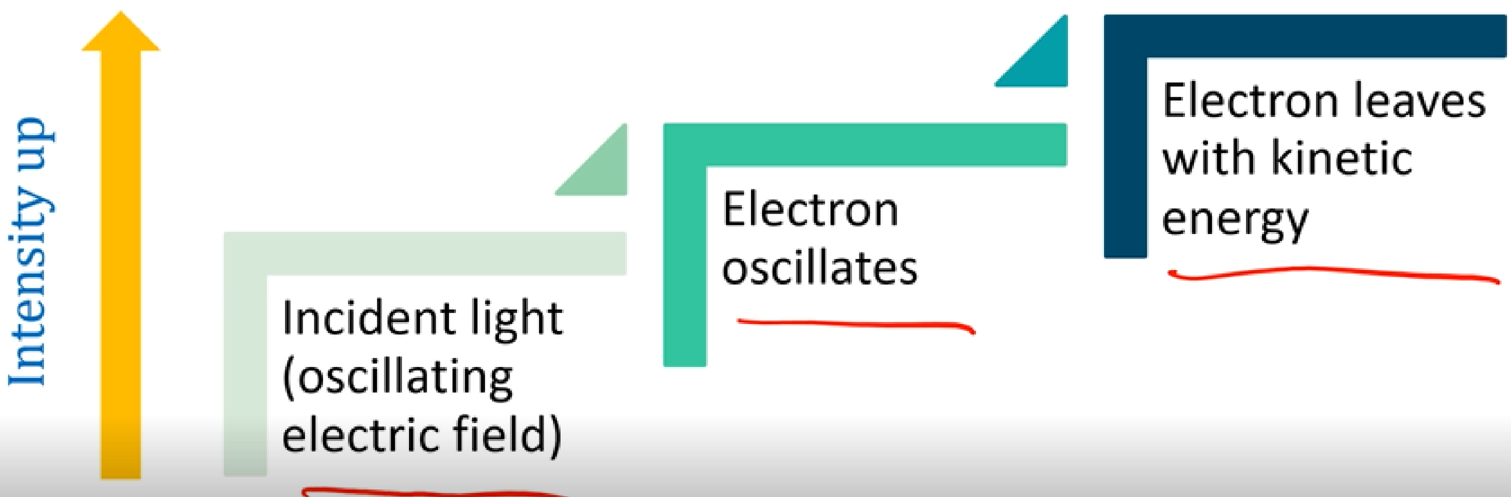
Photoelectric Effect (光电效应)

In 1887 discovered by Hertz;

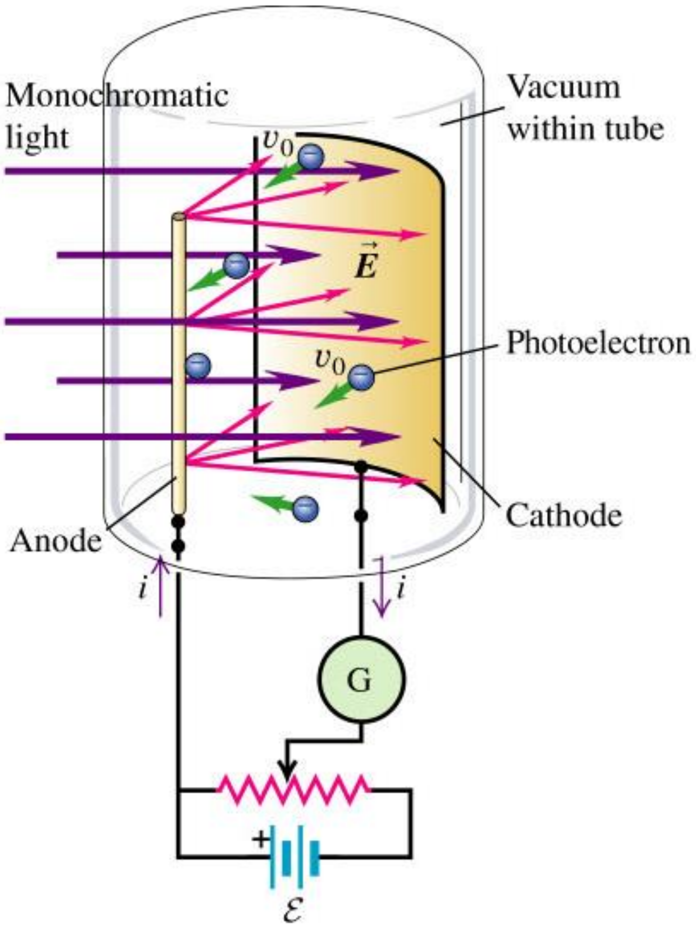
In 1900 Lenard "triggering hypothesis":
No energy transfer when light hits electrons. Light is just like a trigger.

Classically, according to Maxwell equations:

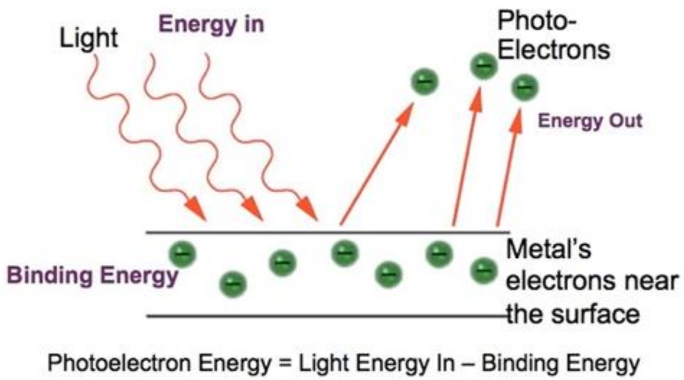


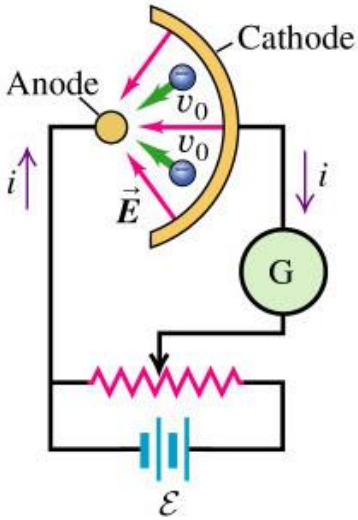


Photoelectric effect



(a)





(b) Overhead view with $\vec{E}$ field reversed

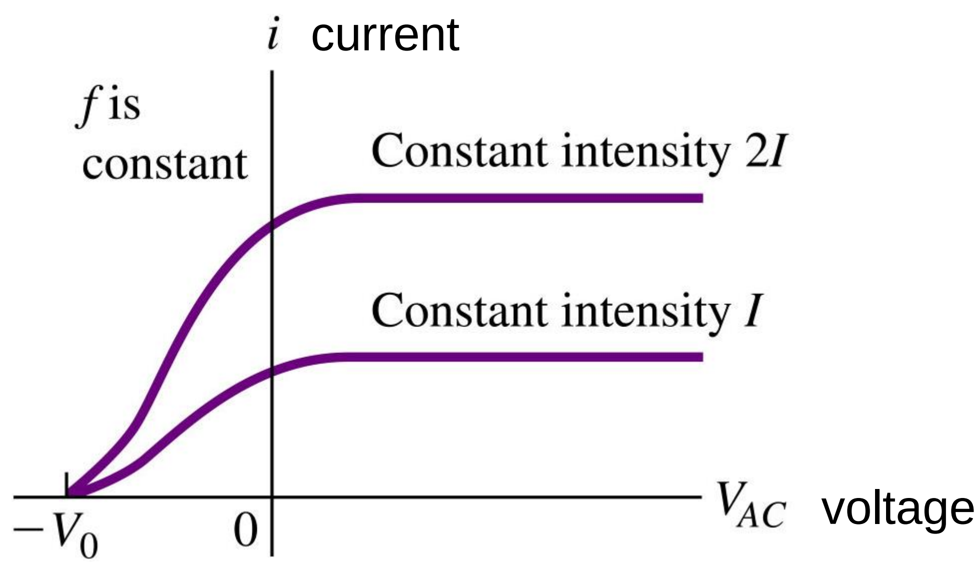
First Photoelectric Experiment

We adjust the potential difference V by moving the sliding contact in Fig. 38-1 so that collector C is slightly negative with respect to target T. This potential difference acts to slow down the ejected electrons. We then vary V until it reaches a certain value, called the **stopping potential** V_{stop}, at which point the reading of meter A has just dropped to zero. When $V = V_{stop}$, the most energetic ejected electrons are turned back just before reaching the collector. Then K_{max}, the kinetic energy of these most energetic electrons, is

$$K_{max} = eV_{stop}, \tag{38-4}$$

where e is the elementary charge.

K_{max} does not depend on the intensity of the light source!



Second Photoelectric Experiment

Now we vary the frequency f of the incident light and measure the associated stopping potential V_{stop}. Figure 38-2 is a plot of V_{stop} versus f. Note that the photoelectric effect does not occur if the frequency is below a certain **cutoff frequency** f_0 or, equivalently, if the wavelength is greater than the corresponding **cutoff wavelength** $\lambda_0 = c/f_0$. This is so *no matter how intense the incident light is.*

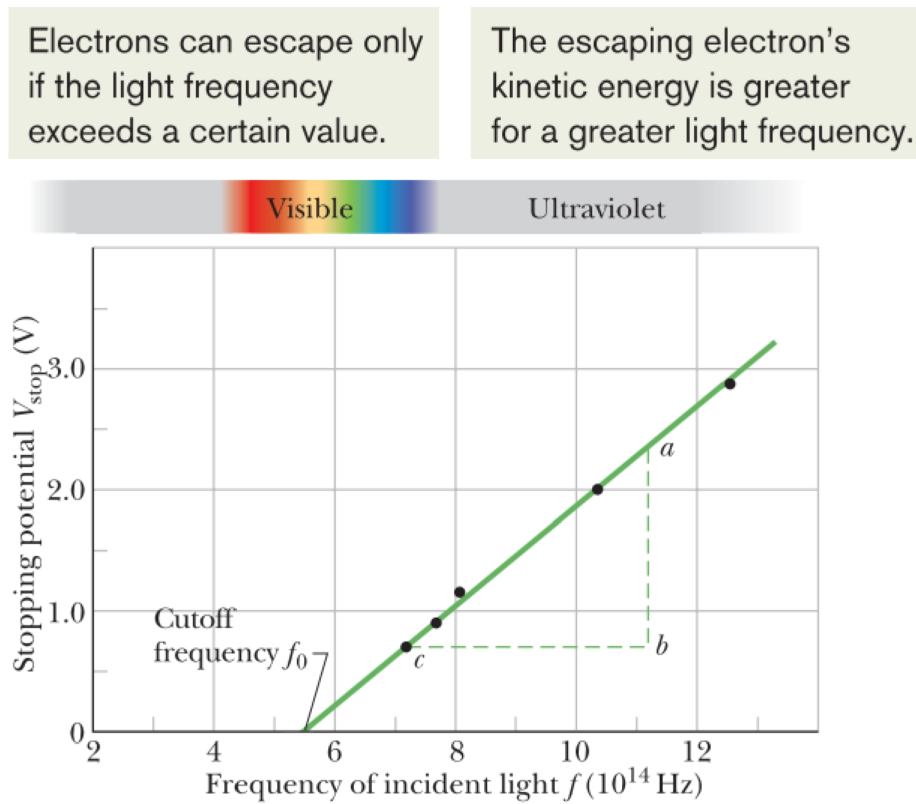


Figure 38-2 The stopping potential V_{stop} as a function of the frequency f of the incident light for a sodium target T in the apparatus of Fig. 38-1. (Data reported by R. A. Millikan in 1916.)

minimum escape energy (material dependent)
or called work function

The Photoelectric Equation

Einstein summed up the results of such photoelectric experiments in the equation

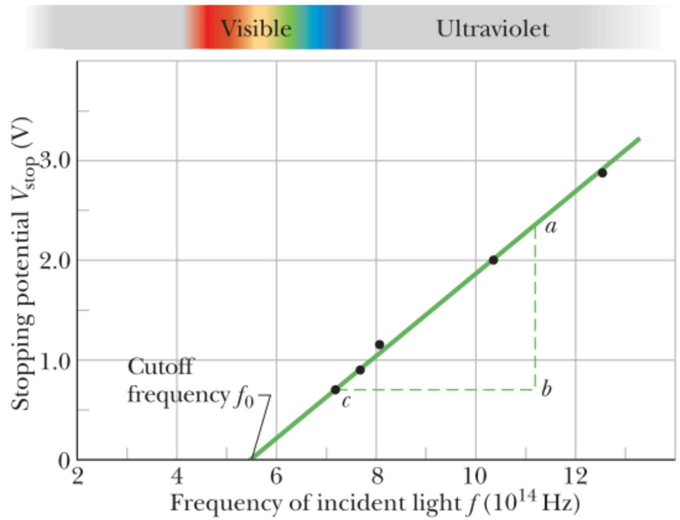
$$hf = K_{\text{max}} + \Phi \quad \text{(photoelectric equation).} \tag{38-5}$$

energy of photon

Let us rewrite Eq. 38-5 by substituting for K_{max} from Eq. 38-4 ($K_{\text{max}} = eV_{\text{stop}}$). After a little rearranging we get

$$V_{\text{stop}} = \left(\frac{h}{e}\right) f - \frac{\Phi}{e}. \tag{38-6}$$

$$V_{\text{stop}} = \left(\frac{h}{e}\right) f - \frac{\Phi}{e}. \tag{38-6}$$



line, as it is in Fig. 38-2. Further, the slope of that straight line should be h/e. As a check, we measure ab and bc in Fig. 38-2 and write

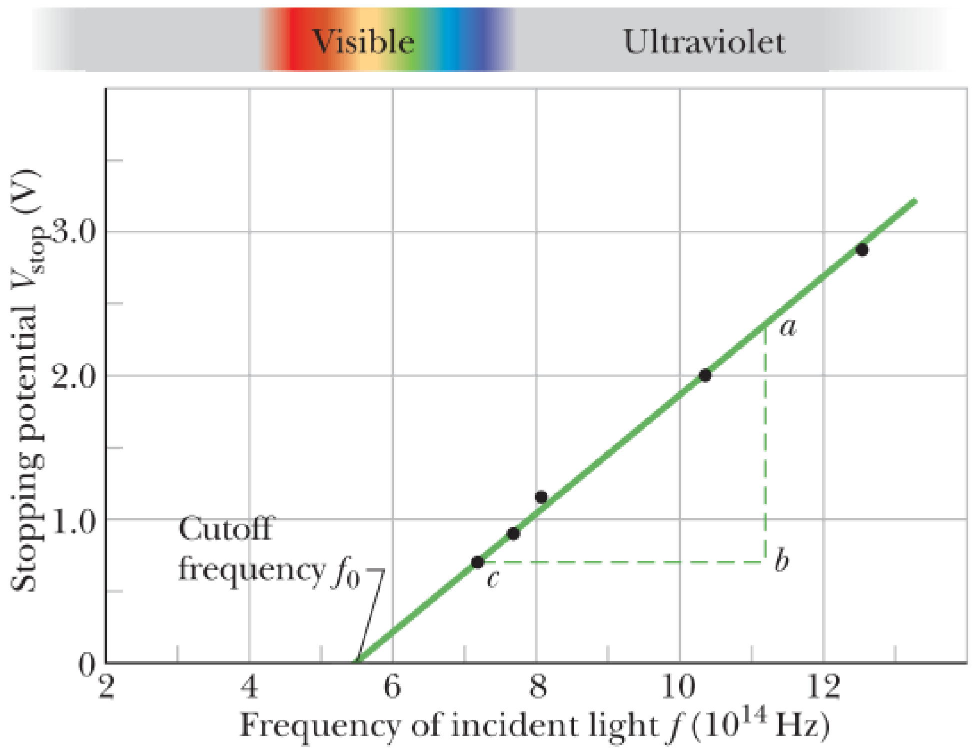
$$\frac{h}{e} = \frac{ab}{bc} = \frac{2.35 \text{ V} - 0.72 \text{ V}}{(11.2 \times 10^{14} - 7.2 \times 10^{14}) \text{ Hz}}$$

$$= 4.1 \times 10^{-15} \text{ V} \cdot \text{s}.$$

Multiplying this result by the elementary charge e, we find

$$h = (4.1 \times 10^{-15}\ \mathrm{V \cdot s})(1.6 \times 10^{-19}\ \mathrm{C}) = 6.6 \times 10^{-34}\ \mathrm{J \cdot s},$$

which agrees with values measured by many other methods.



Photoelectric effect

Key Ideas

- When light of high enough frequency illuminates a metal surface, electrons can gain enough energy to escape the metal by absorbing photons in the illumination, in what is called the photoelectric effect.
- The conservation of energy in such an absorption and escape is written as

$$hf = K_{max} + \Phi,$$

where hf is the energy of the absorbed photon, K_{max} is the kinetic energy of the most energetic of the escaping electrons, and Φ (called the work function) is the least energy required by an electron to escape the electric forces holding electrons in the metal.

- If $hf = \Phi$, electrons barely escape but have no kinetic energy and the frequency is called the cutoff frequency f_0.
- If $hf < \Phi$, electrons cannot escape.

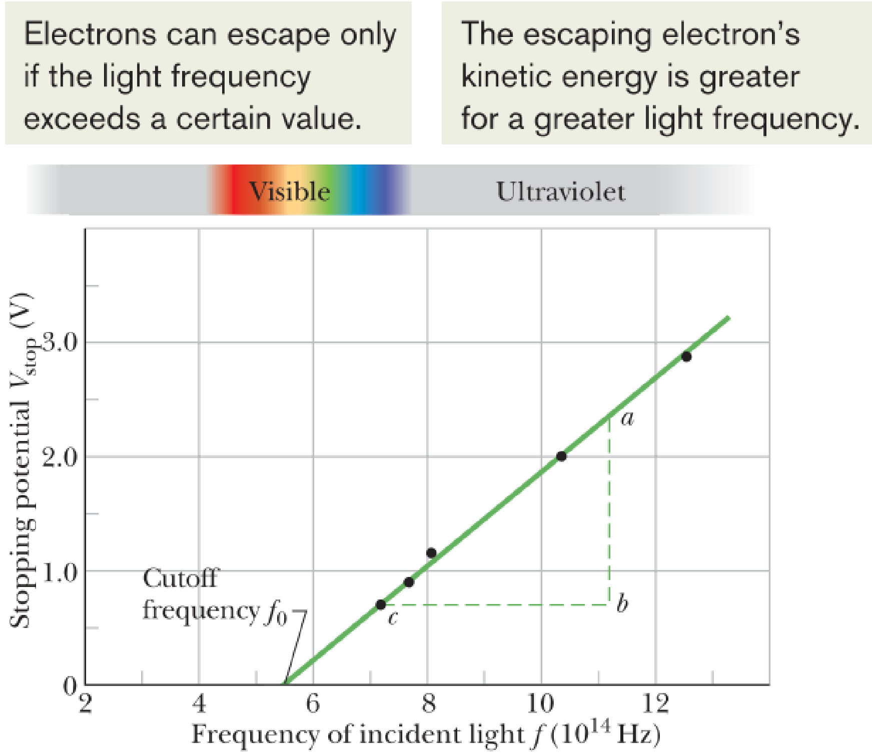


Figure 38-2 The stopping potential V_{stop} as a function of the frequency f of the incident light for a sodium target T in the apparatus of Fig. 38-1. (Data reported by R. A. Millikan in 1916.)

How to prove that photon has **momentum** and energy from experiment

Compton effect

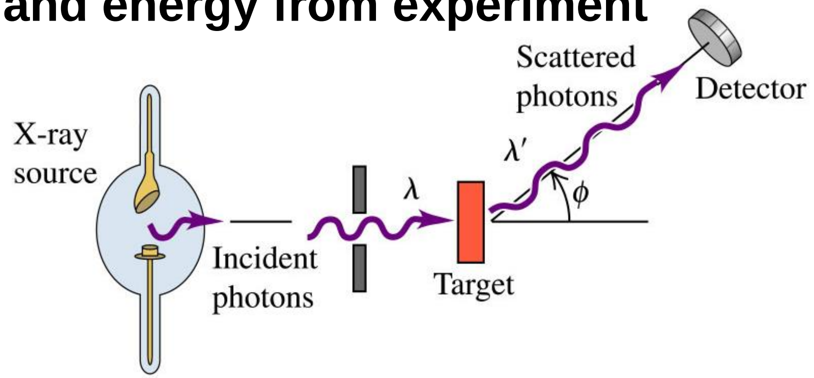


Figure 38-4 shows his results. Although there is only a single wavelength ($\lambda = 71.1$ pm) in the incident x-ray beam, we see that the scattered x rays contain a range of wavelengths with two prominent intensity peaks. One peak is centered about the incident wavelength λ, the other about a wavelength λ' that is longer than λ by an amount $\Delta\lambda$, which is called the **Compton shift.** The value of the Compton shift varies with the angle at which the scattered x rays are detected and is greater for a greater angle.

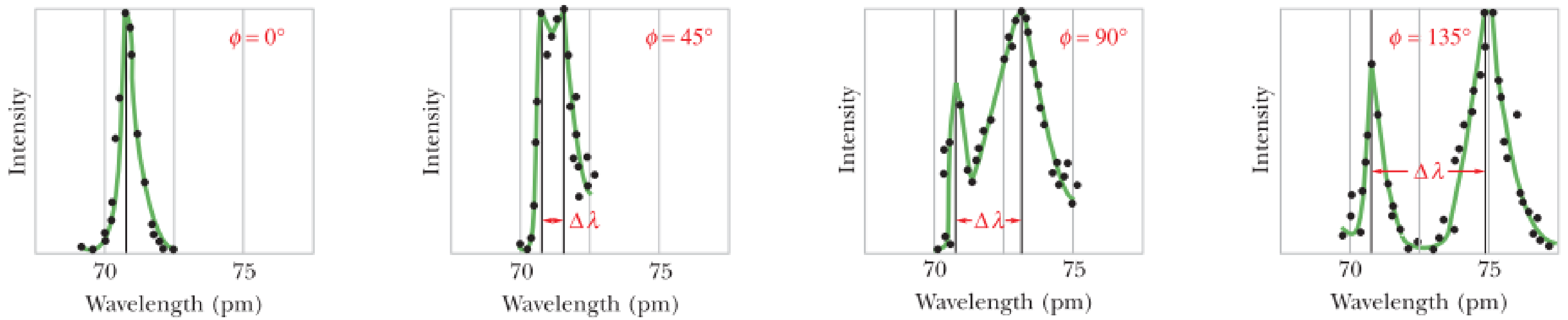
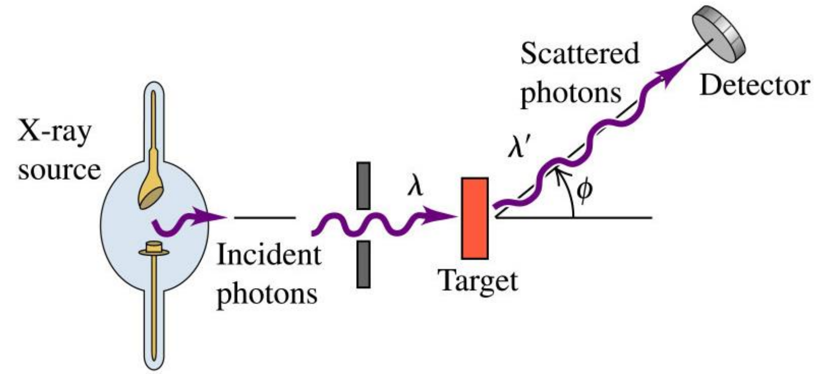


Figure 38-4 Compton's results for four values of the scattering angle ϕ. Note that the Compton shift $\Delta\lambda$ increases as the scattering angle increases.

From classical theory, we should expect no wavelength shift!

Compton effect



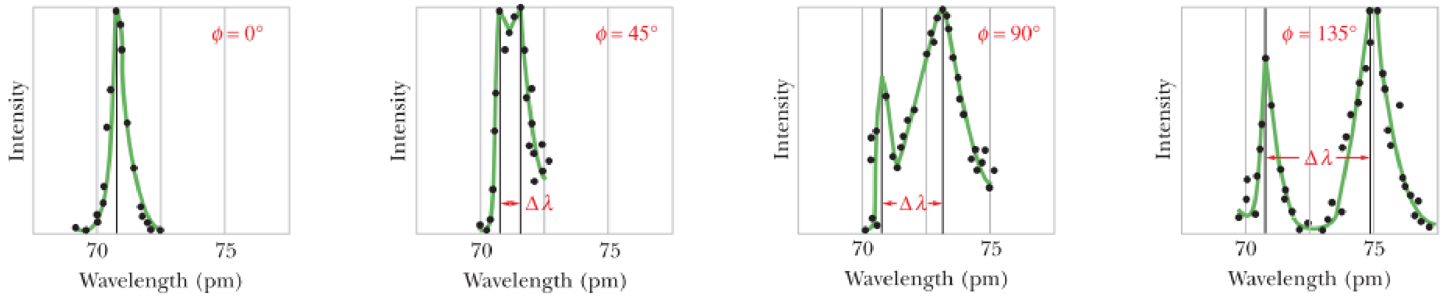
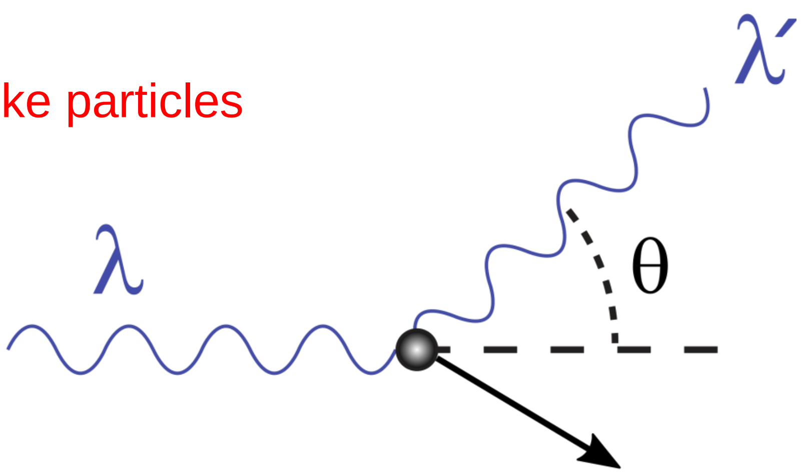
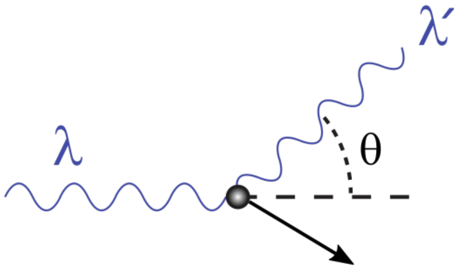


Figure 38-4 Compton's results for four values of the scattering angle ϕ. Note that the Compton shift $\Delta\lambda$ increases as the scattering angle increases.

From quantum theory, photons are like particles



Compton scattering



$$hf = hf' + K, \qquad K = mc^2(\gamma - 1) \qquad \gamma = \frac{1}{\sqrt{1-(v/c)^2}}.$$

in which hf is the energy of the incident x-ray photon, hf' is the energy of the scattered x-ray photon, and K is the kinetic energy of the recoiling electron.

In terms of wavelengths, conservation of energy becomes

$$\boxed{\frac{h}{\lambda} = \frac{h}{\lambda'} + mc(\gamma - 1).}$$

we need this result later

recall: $f = c/\lambda$

Compton scattering

Photons Have Momentum

In 1916, Einstein extended his concept of light quanta (photons) by proposing that a quantum of light has linear momentum. For a photon with energy hf, the magnitude of that momentum is

$$p = \frac{hf}{c} = \frac{h}{\lambda} \quad \text{(photon momentum)}, \tag{38-7}$$

Compton scattering

$$p = \frac{hf}{c} = \frac{h}{\lambda} \quad \text{(photon momentum)},$$

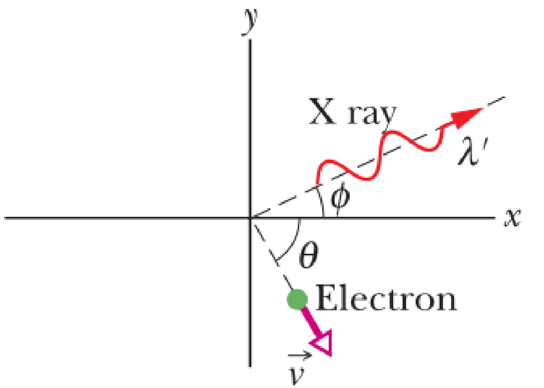
combining these results:

$p = \gamma m v$ (electron momentum)

$$\frac{h}{\lambda} = \frac{h}{\lambda'} \cos\phi + \gamma m v \cos\theta \quad (x \text{ axis})$$

$$0 = \frac{h}{\lambda'} \sin\phi - \gamma m v \sin\theta \quad (y \text{ axis}).$$

$$\frac{h}{\lambda} = \frac{h}{\lambda'} + mc(\gamma - 1).$$



$$\Delta\lambda = \frac{h}{mc}(1 - \cos\phi) \quad \text{(Compton shift)}. \qquad (38\text{-}11)$$

where $\Delta\lambda \; (= \lambda' - \lambda)$.

https://qyumath.github.io/2022/06/30/a226301/

Chapter 38

- **Electron's Double-Slit Experiment**
- **Photons**
- **Matter Wave**
- **Tunneling Through a Potential Barrier**

The Nobel Prize in Physics 1929
Louis de Broglie

Share this: 9

The Nobel Prize in Physics 1929

我的主要结果发表于博士论文 (1924)
还有，我之前是读「历史」的

Prince Louis-Victor Pierre Raymond de Broglie
Prize share: 1/1

The Nobel Prize in Physics 1929 was awarded to Louis de Broglie *"for his discovery of the wave nature of electrons"*.

de Broglie Wave

de Broglie wanted to make the electron behave like a "quanta".
Here is what he did:

Electron must have an intrinsic frequency as

$$E = m_e c^2 = hf \quad \longrightarrow \quad f = m_e c^2/h$$

Frequency is the oscillation period of *something*. What is oscillating?

If electron moves at v_e, a wave comes along (due to relativity) with speed

$$v = c^2/v_e$$

Think why the wave speed can be over c?

So the wavelength

$$\lambda = \frac{v}{f} = \frac{c^2/v_e}{m_e c^2/h} = \frac{h}{m_e v_e} = \frac{h}{p}$$

波速分为群速度和相速度,经典意义下的波在无色散介质中传播的波速是指相速度；在量子力学中所讨论的光子、电子波的速度是指群速度。根据量子力学，德布罗意波代表一个粒子的动量本征态，然而动量本征态是离散态不是束缚态，也就说在实际中是不存在的状态，动量本征态的叠加才是实际存在的状态。所以其速度超光速不会导致实际的对相对论的违背。

Electrons and Matter Waves

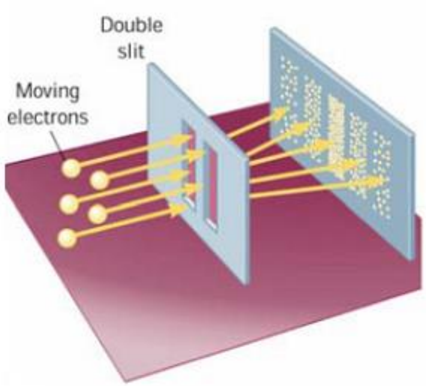
In 1924, French physicist Louis de Broglie made the following appeal to symmetry: A beam of light is a wave, but it transfers energy and momentum to matter only at points, via photons. Why can't a beam of particles have the same properties? That is, why can't we think of a moving electron—or any other particle—as a **matter wave** that transfers energy and momentum to other matter at points?

$$p = \frac{hf}{c} = \frac{h}{\lambda} \quad \text{(photon momentum)}, \tag{38-7}$$

$$\lambda = \frac{h}{p} \quad \text{(de Broglie wavelength)}, \tag{38-17}$$

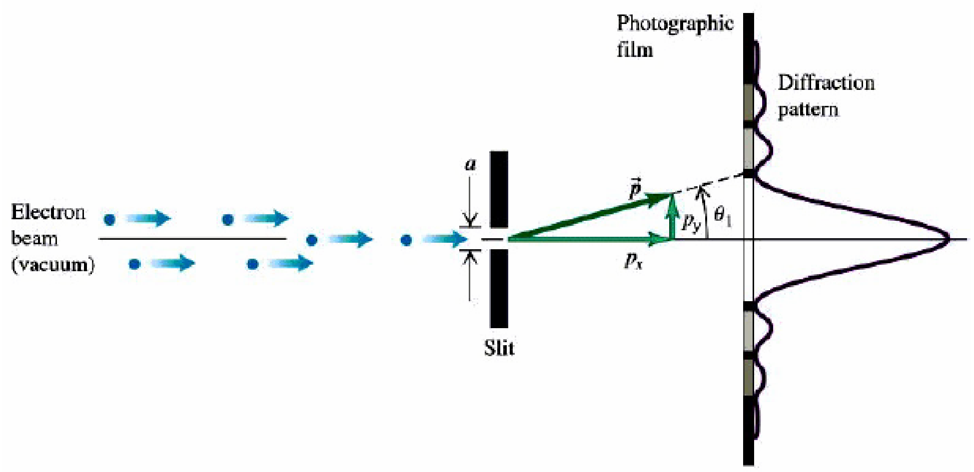
applicable to anything, matter or light

Interference and Diffraction of matter wave



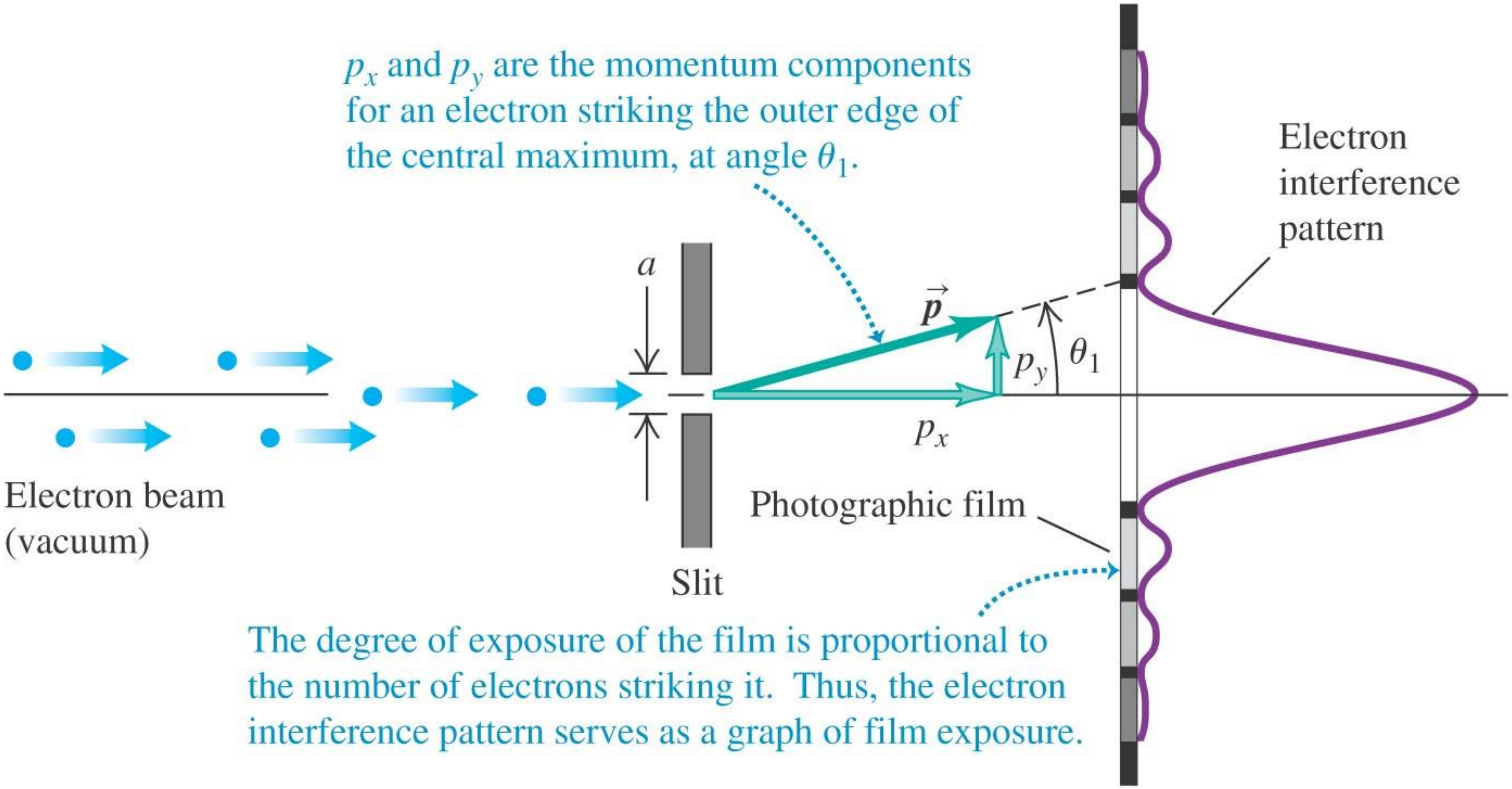
Single slit diffraction experiment:

- A narrow beam of electrons that have very nearly the same speed pass through a slit.
- A diffraction pattern can be observed on a photographic film.



- Consider a beam of electrons all having the same energy and all incident on a double-slit barrier.
- The detector is designed to detect electrons at different positions for a sufficiently long period of time.
- If electrons behaved as classic particles, there would be no interference pattern on the screen.

Single slit experiment with matter wave



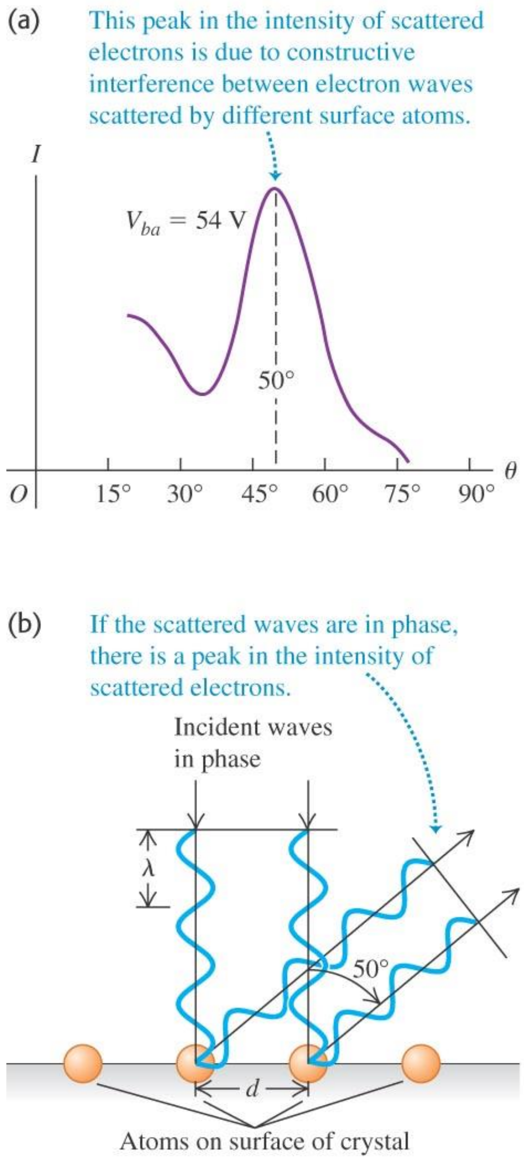
- The frequency f of the *de Broglie wave* is also related to the particle's energy E in the same way as for a photon, namely

$$E = hf$$

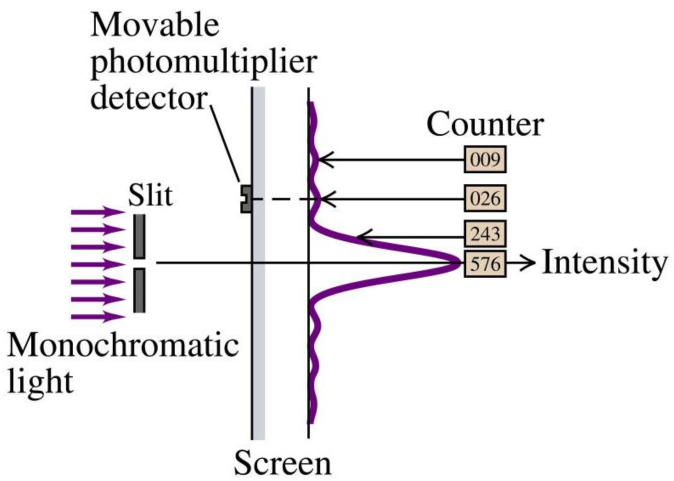
- Experimental Confirmation of de Broglie's Wave Hypothesis
  - Davisson-Germer experiment

    Electron diffraction by nickel crystal

    Sharp maxima in the intensity of the reflected electron beam at specific angles confirmed the wave nature of electrons.



Wave Particle Duality



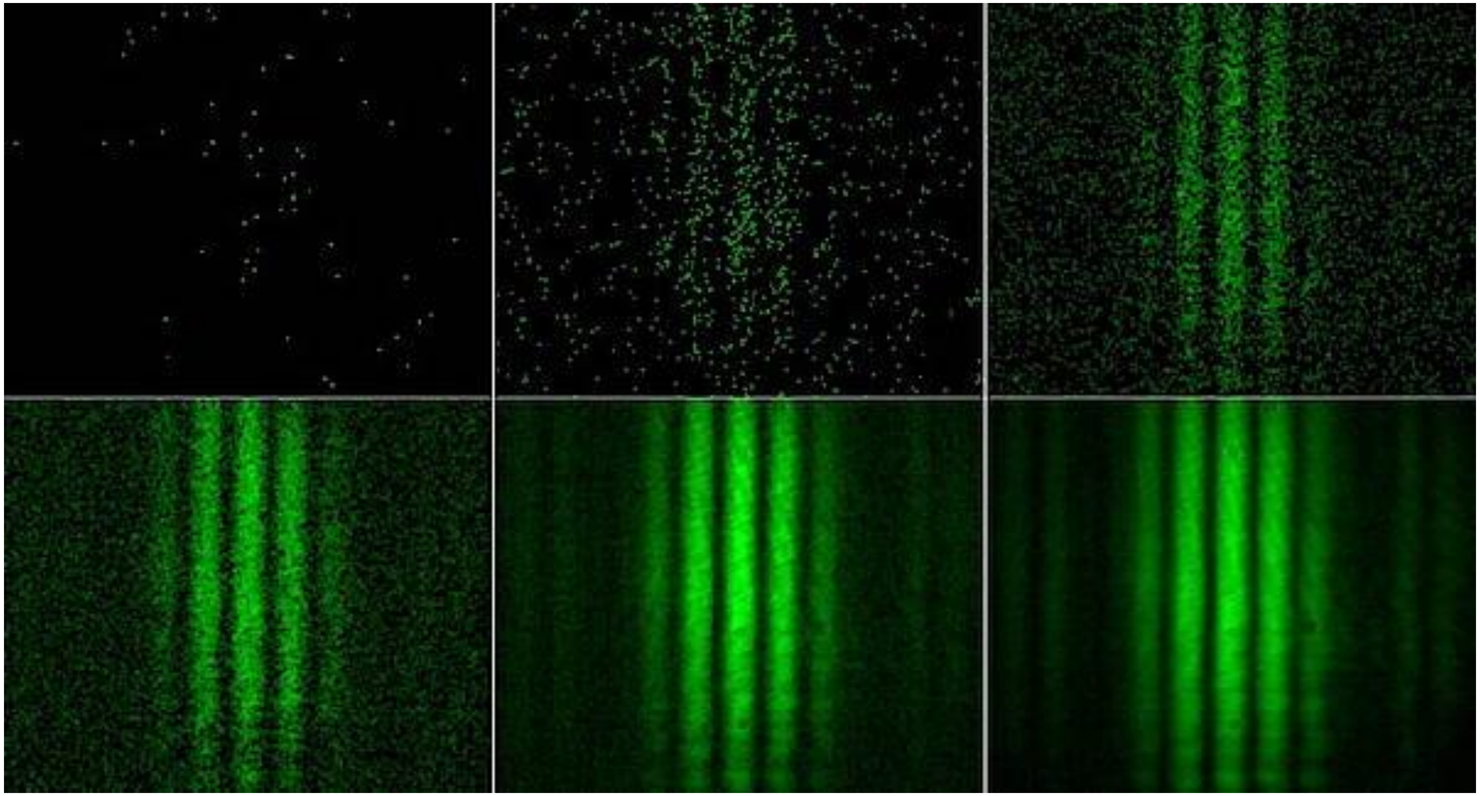
The nature of light

- Light has *wave-light properties* because it shows interference, diffraction and polarization effects.

- Yet the *black body radiation* behavior, *photoelectric effect*, and *Compton effect* suggest that *light behaves like particles* (like protons and electrons).

The Single-Photon Version

A single-photon version of the double-slit experiment was first carried out by G. I. Taylor in 1909 and has been repeated many times since. It differs from the standard version in that the light source in the Taylor experiment is so extremely feeble that it emits only one photon at a time, at random intervals. Astonishingly, interference fringes still build up on screen *C* if the experiment runs long enough (several months for Taylor's early experiment).



Example

Calculate the de Broglie wavelength for (a) an electron ($m=9.11\times10^{-31}$kg) moving at 1.00×10^{7}m/s, and (b) a stone of mass 50 g thrown with a speed of 40 m/s.

Solution

(a) $$\lambda = \frac{h}{p} = \frac{h}{mv} = \frac{6.63\times10^{-34}\,J\cdot s}{(9.11\times10^{-31}\,kg)(1.00\times10^{7}\,m/s)} = 7.28\times10^{-11}\,m$$

(b) $$\lambda = \frac{h}{mv} = \frac{6.63\times10^{-34}\,J\cdot s}{(0.05kg)(40m/s)} = 3.3\times10^{-34}\,m$$

The wavelength of the stone is so short that we do not observe the wave behavior of the stone.

Wave-Particle Duality (波粒二象性)

1. Observation (or measurement) destroys interference pattern!

New name: **collapse (塌缩)** of wave function

A quantum state (like the electron passing the slit), once measured, will collapse to one of its eigenstates (at slit 1 or slit 2) with some probability, and back to the classical world. Before measurement, it is in a "superposition" of its eigenstates. We can never predict its exact behavior, but only know the probability of "collapsing" to one eigenstate. **World is indeterministic.**

2. How to understand wave-particle duality?

Complementarity principle (互补原理)

For the wave-particle duality, it is impossible to observe both the wave and particle aspects simultaneously. Together, however, they present a fuller description than either of the two taken alone.

Niers Bohr (1885 – 1962) Danish physicist; Nobel Prize in 1922.
Leader of the quantum evolution.

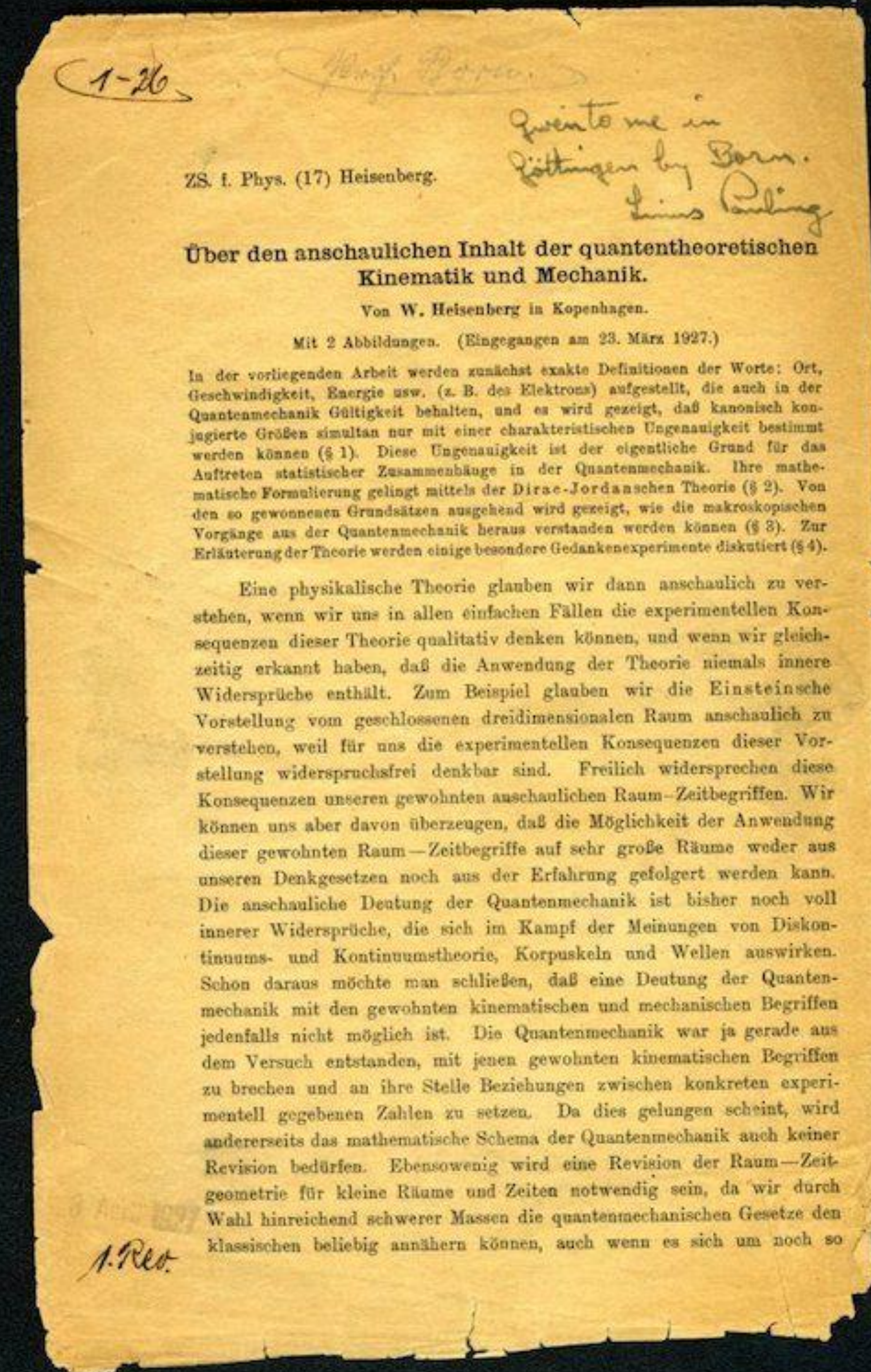
Heisenberg

Heisenberg's Uncertainty Princ

Our inability to predict the position of
energy, as indicated by Fig. 38-13, is ou
principle, proposed in 1927 by Germ
that measured values cannot be assign
of a particle simultaneously with unlim

In terms of $\hbar = h/2\pi$ (called "h-ba

$$\Delta x \cdot \Delta p_x \geq \hbar$$
$$\Delta y \cdot \Delta p_y \geq \hbar \quad \text{(Heisenb}$$
$$\Delta z \cdot \Delta p_z \geq \hbar$$



1-26

Given to me in Göttingen by Born. Linus Pauling

ZS. f. Phys. (17) Heisenberg.

Über den anschaulichen Inhalt der quantentheoretischen Kinematik und Mechanik.

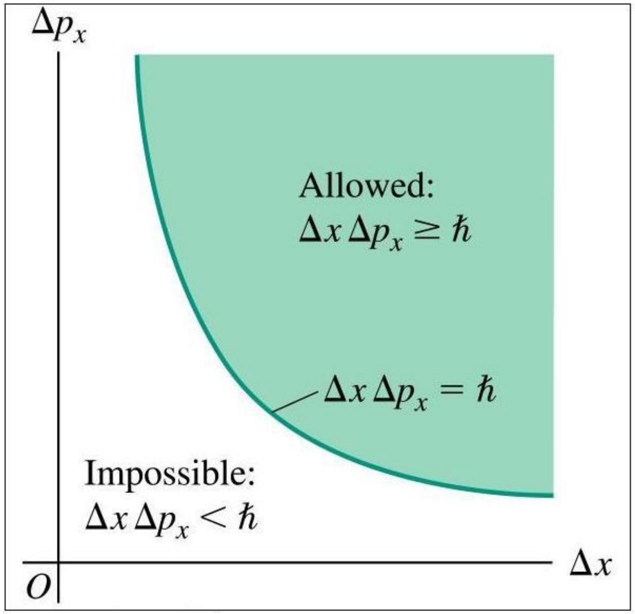
Von W. Heisenberg in Kopenhagen.

Mit 2 Abbildungen. (Eingegangen am 23. März 1927.)

In der vorliegenden Arbeit werden zunächst exakte Definitionen der Worte: Ort, Geschwindigkeit, Energie usw. (z. B. des Elektrons) aufgestellt, die auch in der Quantenmechanik Gültigkeit behalten, und es wird gezeigt, daß kanonisch konjugierte Größen simultan nur mit einer charakteristischen Ungenauigkeit bestimmt werden können (§ 1). Diese Ungenauigkeit ist der eigentliche Grund für das Auftreten statistischer Zusammenhänge in der Quantenmechanik. Ihre mathematische Formulierung gelingt mittels der Dirac-Jordanschen Theorie (§ 2). Von den so gewonnenen Grundsätzen ausgehend wird gezeigt, wie die makroskopischen Vorgänge aus der Quantenmechanik heraus verstanden werden können (§ 3). Zur Erläuterung der Theorie werden einige besondere Gedankenexperimente diskutiert (§ 4).

Eine physikalische Theorie glauben wir dann anschaulich zu verstehen, wenn wir uns in allen einfachen Fällen die experimentellen Konsequenzen dieser Theorie qualitativ denken können, und wenn wir gleichzeitig erkannt haben, daß die Anwendung der Theorie niemals innere Widersprüche enthält. Zum Beispiel glauben wir die Einsteinsche Vorstellung vom geschlossenen dreidimensionalen Raum anschaulich zu verstehen, weil für uns die experimentellen Konsequenzen dieser Vorstellung widerspruchsfrei denkbar sind. Freilich widersprechen diese Konsequenzen unseren gewohnten anschaulichen Raum-Zeitbegriffen. Wir können uns aber davon überzeugen, daß die Möglichkeit der Anwendung dieser gewohnten Raum—Zeitbegriffe auf sehr große Räume weder aus unseren Denkgesetzen noch aus der Erfahrung gefolgert werden kann. Die anschauliche Deutung der Quantenmechanik ist bisher noch voll innerer Widersprüche, die sich im Kampf der Meinungen von Diskontinuums- und Kontinuumstheorie, Korpuskeln und Wellen auswirken. Schon daraus möchte man schließen, daß eine Deutung der Quantenmechanik mit den gewohnten kinematischen und mechanischen Begriffen jedenfalls nicht möglich ist. Die Quantenmechanik war ja gerade aus dem Versuch entstanden, mit jenen gewohnten kinematischen Begriffen zu brechen und an ihre Stelle Beziehungen zwischen konkreten experimentell gegebenen Zahlen zu setzen. Da dies gelungen scheint, wird andererseits das mathematische Schema der Quantenmechanik auch keiner Revision bedürfen. Ebensowenig wird eine Revision der Raum—Zeitgeometrie für kleine Räume und Zeiten notwendig sein, da wir durch Wahl hinreichend schwerer Massen die quantenmechanischen Gesetze den klassischen beliebig annähern können, auch wenn es sich um noch so

1. Rec.



Example

The speed of an electron is measured to be $5.00 \text{X} 10^3$ m/s to an accuracy of 0.00300%. Find the minimum uncertainty in determining the position of this electron.

Solution

- The momentum of the electron

$$p_x = mv = (9.11\times10^{-31}\,kg)(5.00\times10^3\,m/s)$$
$$= 4.56\times10^{-27}\,kg\cdot m/s$$

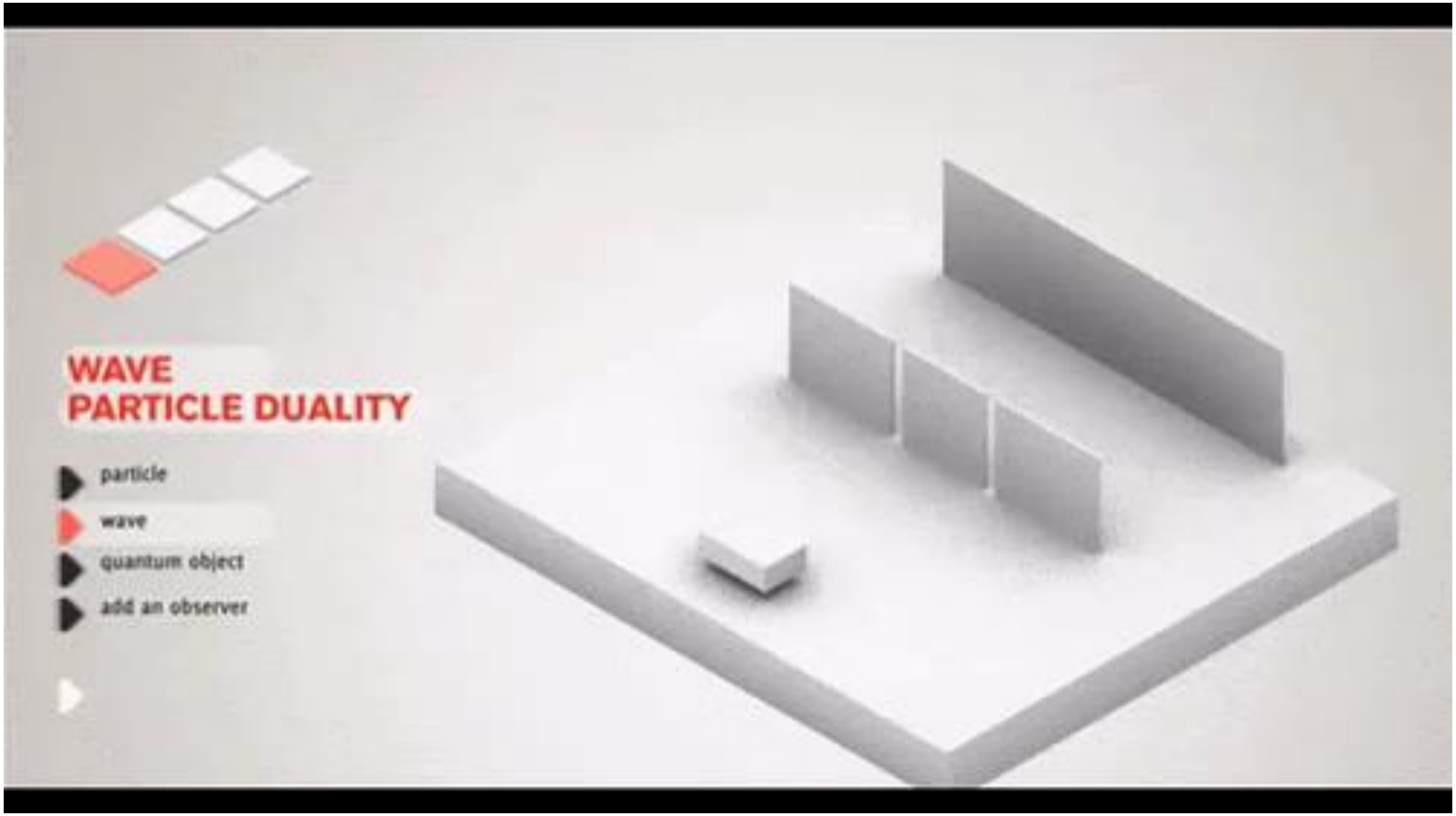
- The uncertainty in p_x

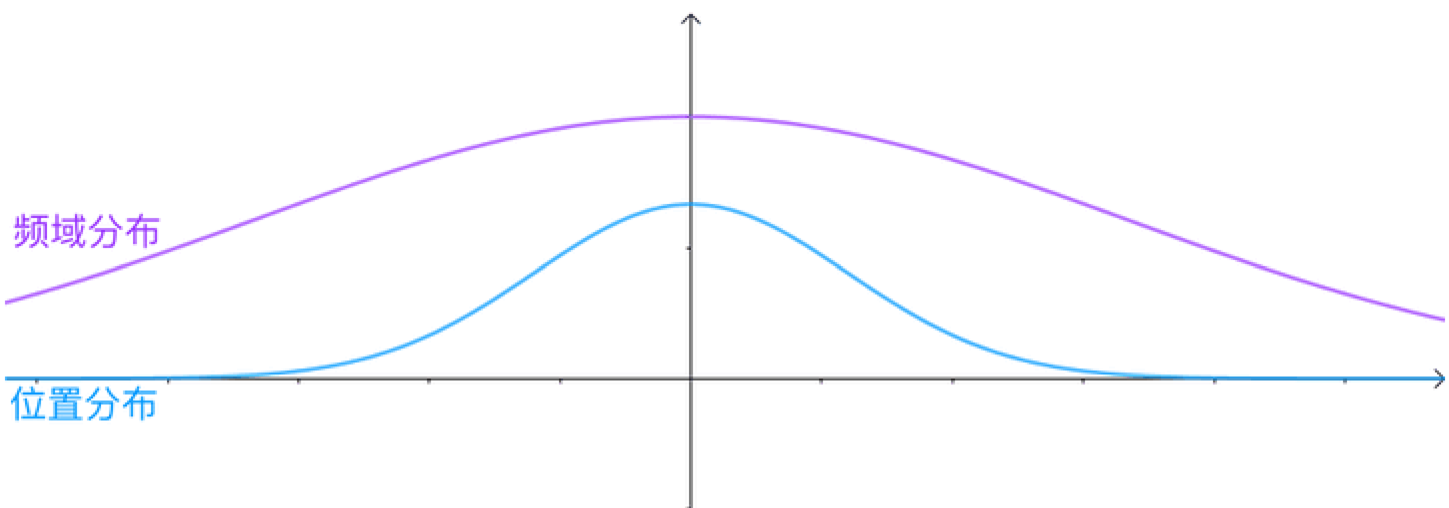
$$\Delta p_x = (0.0000300)(4.56\times10^{-27}\,kg\cdot m/s)$$
$$= 1.37\times10^{-31}\,kg\cdot m/s$$

- Calculate the minimum uncertainty in position by using this value of Δp_x and uncertainty principle

$$\Delta x \Delta p_x \geq \frac{\hbar}{2}$$

$$\Delta x \geq \frac{\hbar}{2\Delta p_x} = \frac{1.05\times10^{-34}\,J\cdot s}{2(1.37\times10^{-31}\,kg\cdot m/s} = 0.383mm$$


WAVE
PARTICLE DUALITY
particle
wave
quantum object
add an observer


频域分布
位置分布

- In quantum mechanics, matter waves are described by the complex-valued wave function Ψ.

$$\Psi = \Psi(x, y, z, t) = \psi(x, y, z)e^{-i\omega t}$$

where imaginary number $i = \sqrt{-1}$

- The absolute square gives the probability of finding a particle at a given point at some instant.

$$|\Psi|^2 = \Psi^*\Psi$$

- Where Ψ^* is the complex conjugate of Ψ.

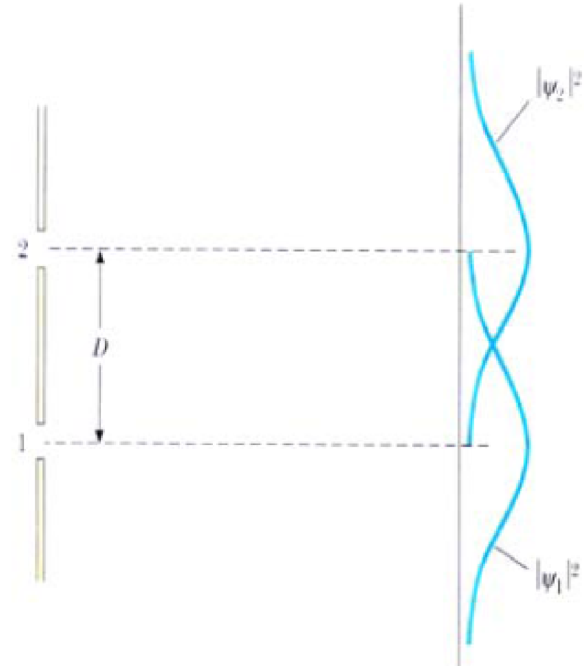
Max Born

The probability of detecting a particle in a small volume centered on a given point in a matter wave is proportional to the value of $|\psi|^2$ at that point.

- If one slit is covered during experiment, the result is a symmetric curve that peaks around the center of the open slit, much like the pattern formed by bullets shot through a hole in armor plate.
- These curves are expressed as:

$$|\psi_1|^2 = \psi_1^* \psi_1$$

$$|\psi_2|^2 = \psi_2^* \psi_2$$



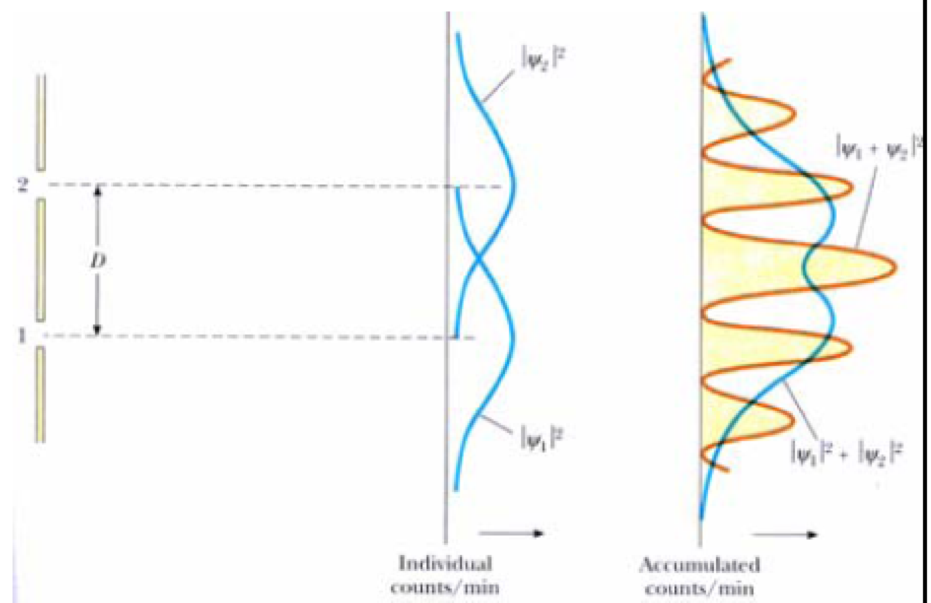
- The accumulated result is simply the sum of the individual results, and the interference pattern lost. This curve can be expressed as: $|\psi_1|^2 + |\psi_2|^2$

- The electron is in superposition state when both slits are open:

$$\psi = \psi_1 + \psi_2$$

- The probability of detecting the electron at the detector is:

$$\psi^2 = |\psi_1 + \psi_2|^2$$



All measurable quantities of a particle, such as its energy and linear momentum, can be derived from a knowledge of $\boldsymbol{\Psi}$

The average position x of the particle, which is often called the expectation value of x, is defined by the equation

$$\langle x \rangle = \int_{-\infty}^{\infty} x|\psi|^2 dx$$

Schrödinger Equation

Every wave has a wave equation. What is the wave equation of the probability wave that (non-relativistic particles must obey)? Schrödinger in 1925 "derived" the equation what is now named after him,

$$i\hbar\frac{\partial}{\partial t}\Psi(\mathbf{r},t) = \left[\frac{-\hbar^2}{2m_e}\nabla^2 + V(\mathbf{r},t)\right]\Psi(\mathbf{r},t)$$

Time-dependent

Nobel Price (1933)

If the quantum state does not depend on time, the equation reduces to

$$\left[\frac{-\hbar^2}{2m_e}\nabla^2 + V(\mathbf{r})\right]\Psi(\mathbf{r}) = E\Psi(\mathbf{r})$$

Time-independent

Schrödinger Equation

$$\frac{d^2\psi}{dx^2} + k^2\psi = 0 \quad \text{(Schrödinger's equation, uniform } U\text{)}, \tag{38-22}$$

The general solution of Eq. 38-22 is

$$\psi(x) = Ae^{ikx} + Be^{-ikx}, \tag{38-24}$$

in which A and B are constants. You can show that this equation is indeed a solution of Eq. 38-22 by substituting it and its second derivative into that equation and noting that an identity results.

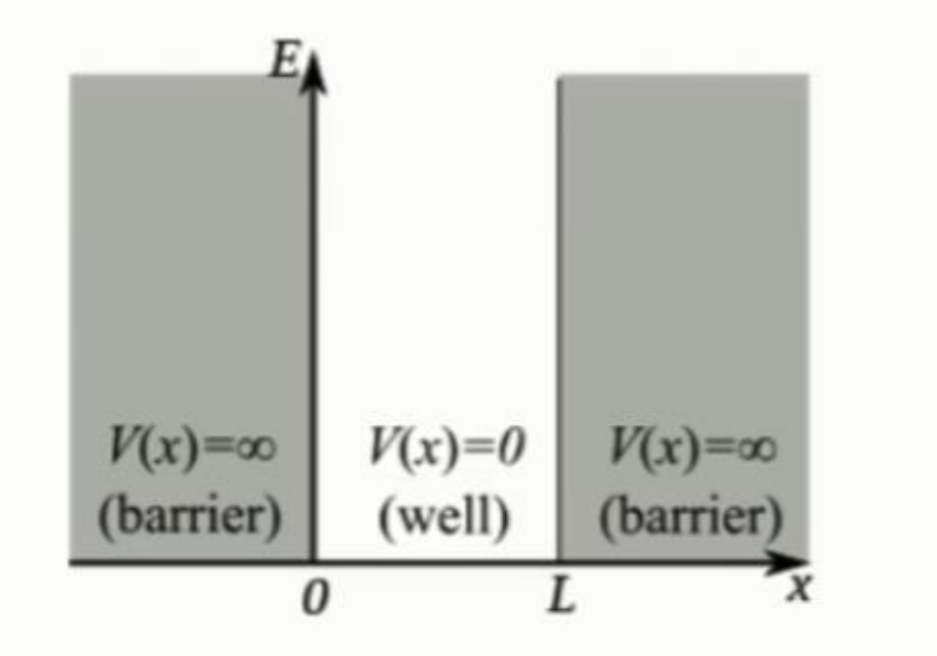
Equation 38-24 is the time-independent solution of Schrödinger's equation. We can assume it is the spatial part of the wave function at some initial time $t = 0$. Given values for E and U, we could determine the coefficients A and B to see how the wave function looks at $t = 0$. Then, if we wanted to see how the wave function evolves with time, we follow the guide of Eq. 38-18 and multiply Eq. 38-24 by the time dependence $e^{-i\omega t}$:

$$\begin{aligned}\Psi(x,t) = \psi(x)e^{-i\omega t} &= (Ae^{ikx} + Be^{-ikx})e^{-i\omega t} \\ &= Ae^{i(kx-\omega t)} + Be^{-i(kx+\omega t)}.\end{aligned} \tag{38-25}$$

(1) A particle in a well of infinite height

- Consider a particle having a potential energy:

$$U(x) = \begin{cases} 0 & 0 \le x \le L \\ \infty & x < 0 \text{ and x} > \text{L} \end{cases}$$



- Schrödinger Equation can be expressed in the form:

$$\frac{d^2\psi}{dx^2} + k^2\psi = 0 \qquad \text{where } k = \frac{\sqrt{2mE}}{\hbar}$$

The wave function must meet following condition:

(i) Wave function must be continuous. That is, the wave function for different regions must join smoothly at the boundaries;

(ii) For $\psi(x, y, z)$ to obey the normalization condition, it is required that $\psi(x, y, z)$ approaches zero as x, y, and z approach $\pm\infty$.

(iii) $\psi(x, y, z)$ must be single-valued, and $d\psi/dx$ must also be continuous for finite values of $U(x)$.

- Solution to the Schrödinger Equation
  - ψ must be zero outside the box and at the walls because the walls are infinitely high and the particle is permanently bound between walls and cannot be found outside.
  - The general solution for a free particle is

$$\begin{aligned}\psi(x) &= A'e^{ikx} + B'e^{-ikx} \\ &= A'(\cos kx + i\sin kx) + B'(\cos kx - i\sin kx) \\ &= A\cos kx + B\sin kx\end{aligned}$$

  - The solution must meet the boundary conditions in the well.

$\psi(x)=0$ at $x = 0$ and $x = L$.

$$\psi(x) = A'e^{ikx} + B'e^{-ikx}$$
$$= A'(\cos kx + i\sin kx) + B'(\cos kx - i\sin kx)$$
$$= A\cos kx + B\sin kx$$

For $\psi(0)=0$,

$$0 = A\cos(k\cdot 0) + B\sin(k\cdot 0) \quad \Rightarrow \quad A = 0$$

$$\therefore \quad \psi(x) = B\sin kx$$

For $\psi(L)=0$,

$$B\sin kL = 0 \quad \Rightarrow \quad kL = \frac{L\sqrt{2mE}}{\hbar} = n\pi \qquad \text{n} = 0, 1, 2, \ldots$$

When n = 0, ψ =0, there is no physical meaning.

For $\psi(0)=0$,

$$0 = A\cos(k\cdot 0) + B\sin(k\cdot 0) \quad \Rightarrow \quad A = 0$$

$$\therefore \quad \psi(x) = B\sin kx$$

For $\psi(L)=0$,

$$B\sin kL = 0 \quad \Rightarrow \quad kL = \frac{L\sqrt{2mE}}{\hbar} = n\pi \qquad \mathrm{n} = 0, 1, 2, \ldots$$

When n = 0, ψ =0, there is no physical meaning.

- Solving for the allowed energies E gives

$$E_n = \left(\frac{h^2}{8mL^2}\right)n^2 \qquad \mathrm{n} = 1, 2, 3, \ldots\ldots$$

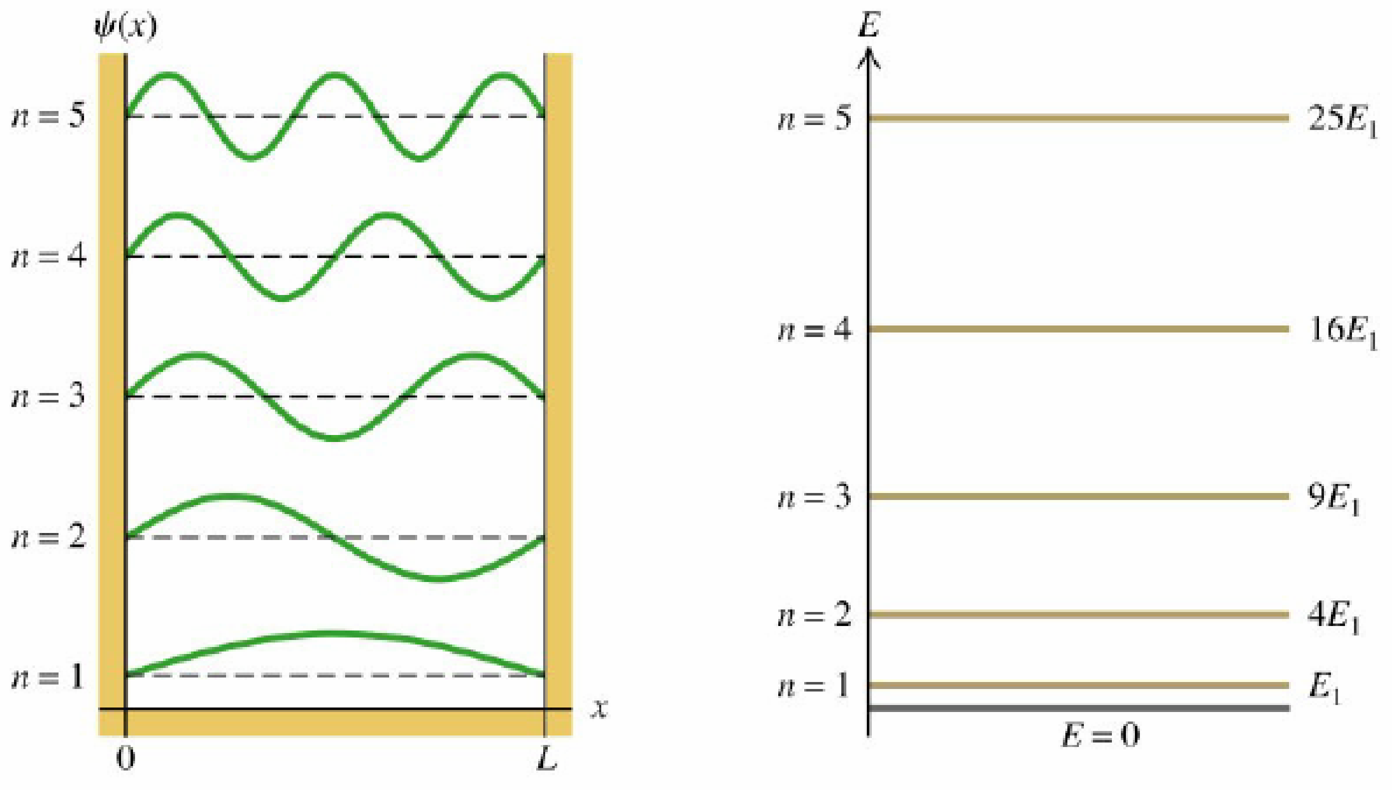
- (i) **The lowest energy level** is $E_1 = \hbar^2\pi^2 / 2mL^2 \neq 0$

  Which is quite different form classic particles.

- (ii) $E_n \propto n^2$, the energy level distribution is not uniform.

- The allowed wave functions are given by

$$\psi_n(x) = B\sin\left(\frac{n\pi x}{L}\right)$$



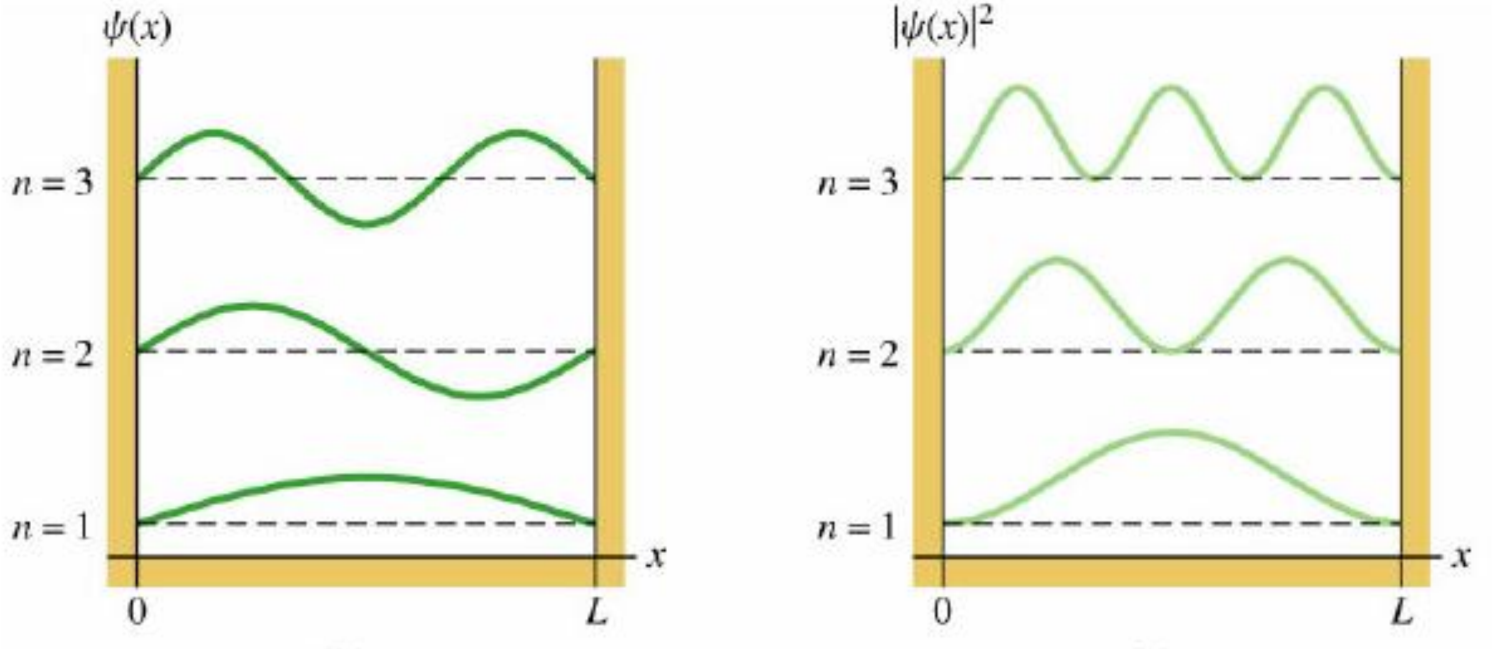
(a) (b)

- The nomalization constant B can be determined by

$$\int_0^L \left|\psi^2(x)\right| dx = \int_0^L B_n^2 \sin^2\left(\frac{n\pi x}{L}\right) dx = 1 \qquad B_n = \sqrt{\frac{2}{L}}$$

$$\therefore \quad \psi_n(x) = \sqrt{\frac{2}{L}} \sin\left(\frac{n\pi x}{L}\right)$$

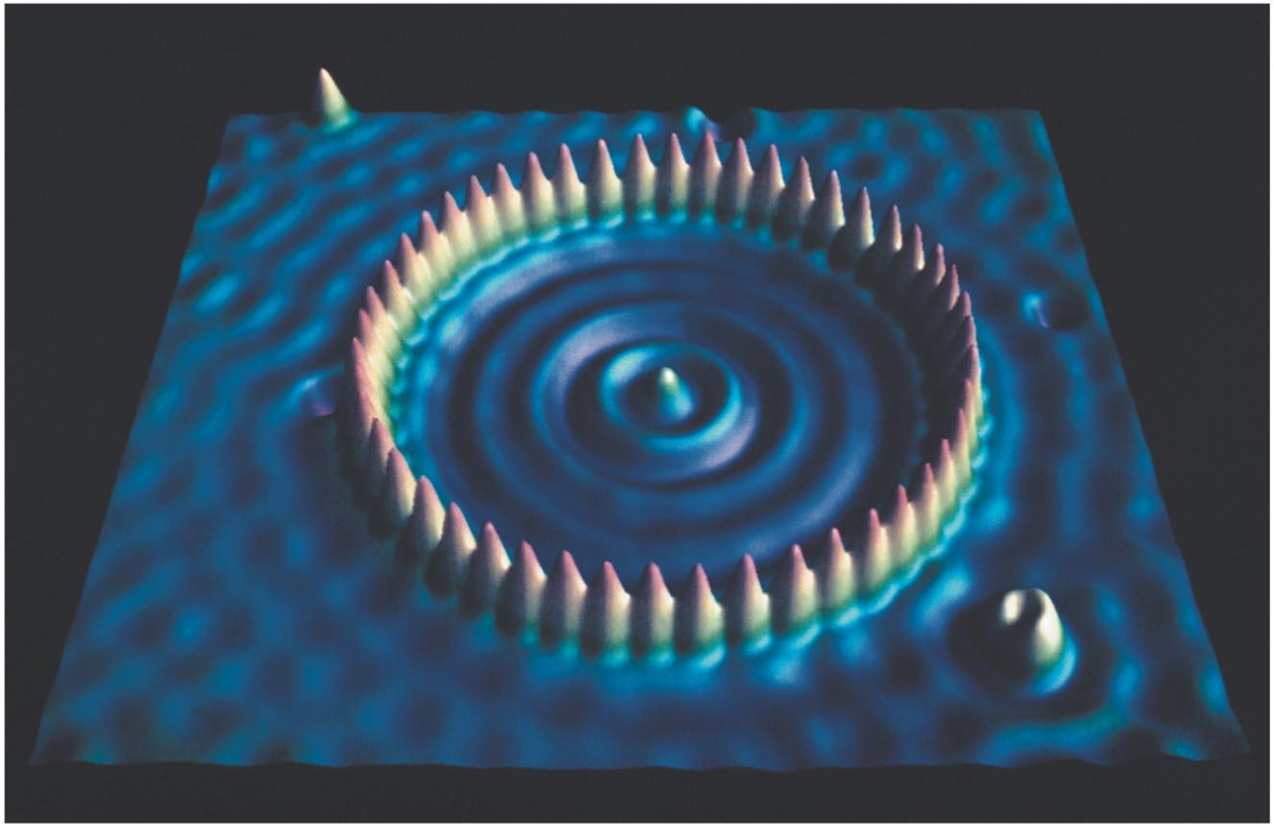
- Probability



(a) (b)

Real atoms in a 2-D well

Surface scientists placed 48 Fe atoms on a Cu surface (this is done at ultra-high vacuum and the image is made with scanning-tunneling electron microscopy).



Chapter 38

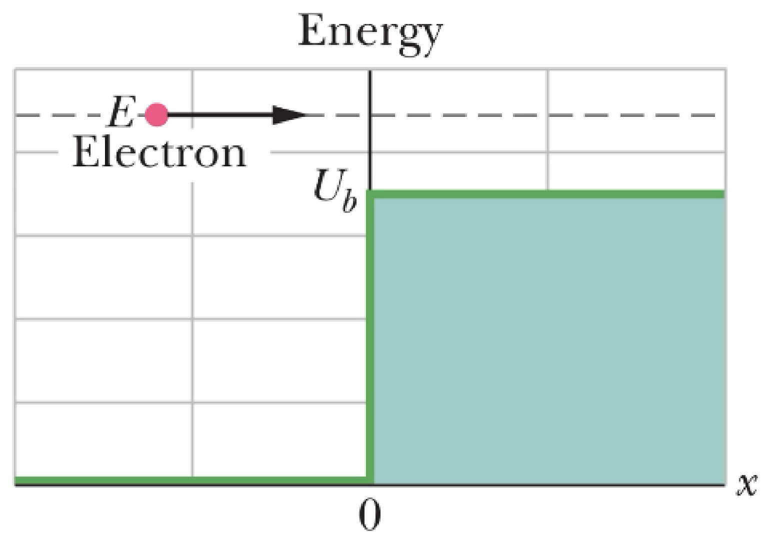
- **Electron's Double-Slit Experiment**
- **Photons**
- **Matter Wave**
- **<u>Tunneling Through a Potential Barrier</u>**

Tunneling Through a Potential Barrier

A beam of nonrelativistic electrons (each energy E) encounter a region with potential energy U_b at $x = 0$. Assume $E > U_b$.

Classically, electrons should all pass through the boundary.

If we apply Schrödinger Equation…



$x < 0$	$x > 0$
$k = \frac{2\pi\sqrt{2mE}}{h}$	$k_b = \frac{2\pi\sqrt{2m(E - U_b)}}{h}$
$\psi_1(x) = Ae^{ikx} + Be^{-ikx}$	$\psi_2(x) = Ce^{ik_b x} + De^{-ik_b x}$

A: incident; B: reflected; C: transmitted; D: 0

Tunneling Through a Potential Barrier

At boundary x =0

$$A + B = C \text{ (matching of values)}$$

Take a derivative at boundary x = 0

$$Ak - Bk = Ck_b \text{ (matching of slopes)}$$

Reflection coefficient

$$R = \frac{|B|^2}{|A|^2}$$

Transmission coefficient

$$T = 1 - R$$

In quantum physics, R is not simply zero (and T is not simply 1) as we assumed in classical physics.

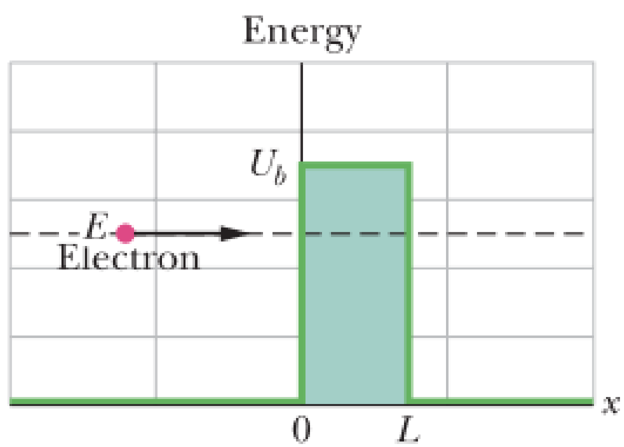
reflected. The transmission coefficient T is approximately

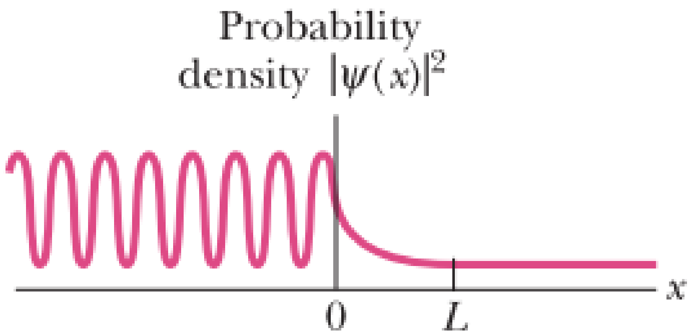
$$T \approx e^{-2bL}, \tag{38-38}$$

in which

$$b = \sqrt{\frac{8\pi^2 m(U_b - E)}{h^2}}, \tag{38-39}$$

and e is the exponential function. Because of the exponential form of Eq. 38-38, the value of T is very sensitive to the three variables on which it depends: particle mass m, barrier thickness L, and energy difference $U_b - E$. (Because we do not include relativistic effects here, E does not include mass energy.)





Example

A 30 eV electron is incident on a barrier whose cross-section is a square of height 40 eV. What is the probability that the electron will tunnel through the barrier if its thickness is (a) 1.0 nm? (b) 0.10nm?

Solution

(a) U-E = (40eV-30eV)=10eV = $1.6\times10^{-18}\ J$

The probability of tunneling through the barrier:

$$T \approx e^{-2CL} = e^{-\frac{2L\sqrt{2m(U-E)}}{\hbar}}$$

$$= \exp\left[2(1.0\times10^{-9}\,m)\frac{\sqrt{2(9.11\times10^{-31}\,kg)(1.6\times10^{-18}\,J)}}{1.054\times10^{-34}\,J\cdot s}\right] = 8.5\times10^{-15}$$

Review & Summary

Light Quanta—Photons An electromagnetic wave (light) is quantized, and its quanta are called *photons*. For a light wave of frequency f and wavelength λ, the energy E and momentum magnitude p of a photon are

$$E = hf \quad \text{(photon energy)} \tag{38-2}$$

and

$$p = \frac{hf}{c} = \frac{h}{\lambda} \quad \text{(photon momentum).} \tag{38-7}$$

Photoelectric Effect When light of high enough frequency falls on a clean metal surface, electrons are emitted from the surface by photon–electron interactions within the metal. The governing relation is

$$hf = K_{\max} + \Phi, \tag{38-5}$$

in which hf is the photon energy, $K_{\max}$ is the kinetic energy of the most energetic emitted electrons, and Φ is the **work function** of the target material—that is, the minimum energy an electron must have if it is to emerge from the surface of the target. If hf is less than Φ, electrons are not emitted.

Compton Shift When x rays are scattered by loosely bound electrons in a target, some of the scattered x rays have a longer wavelength than do the incident x rays. This **Compton shift** (in wavelength) is given by

$$\Delta\lambda = \frac{h}{mc}(1 - \cos\phi), \tag{38-11}$$

in which ϕ is the angle at which the x rays are scattered.

Light Waves and Photons When light interacts with matter, energy and momentum are transferred via photons. When light is in transit, however, we interpret the light wave as a **probability wave,** in which the probability (per unit time) that a photon can be detected is proportional to E_m^2, where E_m is the amplitude of the oscillating electric field of the light wave at the detector.

Ideal Blackbody Radiation As a measure of the emission of thermal radiation by an ideal blackbody radiator, we define the spectral radiancy $S(\lambda)$ in terms of the emitted intensity per unit wavelength at a given wavelength λ. For the Planck radiation law, in which atomic oscillators produce the thermal radiation, we have

$$S(\lambda) = \frac{2\pi c^2 h}{\lambda^5} \frac{1}{e^{hc/\lambda kT} - 1}, \tag{38-14}$$

where h is the Planck constant, k is the Boltzmann constant, and T is the temperature of the radiating surface. Wien's law relates the temperature T of a blackbody radiator and the wavelength λ_{max} at which the spectral radiancy is maximum:

$$\lambda_{max} T = 2898\ \mu\text{m} \cdot \text{K}. \tag{38-15}$$

Matter Waves A moving particle such as an electron or a proton can be described as a **matter wave;** its wavelength (called the **de Broglie wavelength**) is given by $\lambda = h/p$, where p is the magnitude of the particle's momentum.

The Wave Function A matter wave is described by its **wave function** $\Psi(x, y, z, t)$, which can be separated into a space-dependent part $\psi(x, y, z)$ and a time-dependent part $e^{-i\omega t}$. For a particle of mass m moving in the x direction with constant total energy E through a region in which its potential energy is $U(x)$, $\psi(x)$ can be found by solving the simplified **Schrödinger equation:**

$$\frac{d^2\psi}{dx^2} + \frac{8\pi^2 m}{h^2}[E - U(x)]\psi = 0. \tag{38-19}$$

A matter wave, like a light wave, is a probability wave in the sense that if a particle detector is inserted into the wave, the probability that the detector will register a particle during any specified time interval is proportional to $|\psi|^2$, a quantity called the **probability density.**

For a free particle—that is, a particle for which $U(x) = 0$—moving in the x direction, $|\psi|^2$ has a constant value for all positions along the x axis.

Heisenberg's Uncertainty Principle The probabilistic nature of quantum physics places an important limitation on detecting a particle's position and momentum. That is, it is not possible to measure the position $\vec{r}$ and the momentum $\vec{p}$ of a particle simultaneously with unlimited precision. The uncertainties in the components of these quantities are given by

$$\begin{aligned} \Delta x \cdot \Delta p_x &\geq \hbar \\ \Delta y \cdot \Delta p_y &\geq \hbar \\ \Delta z \cdot \Delta p_z &\geq \hbar. \end{aligned} \tag{38-28}$$

Potential Step This term defines a region where a particle's potential energy increases at the expense of its kinetic energy. According to classical physics, if a particle's initial kinetic energy exceeds the potential energy, it should never be reflected by the region. However, according to quantum physics, there is a reflection coefficient R that gives a finite probability of reflection. The probability of transmission is $T = 1 - R$.

Barrier Tunneling According to classical physics, an incident particle will be reflected from a potential energy barrier whose height is greater than the particle's kinetic energy. According to quantum physics, however, the particle has a finite probability of tunneling through such a barrier, appearing on the other side unchanged. The probability that a given particle of mass m and energy E will tunnel through a barrier of height U_b and thickness L is given by the transmission coefficient T:

$$T \approx e^{-2bL}, \tag{38-38}$$

where

$$b = \sqrt{\frac{8\pi^2 m(U_b - E)}{h^2}}. \tag{38-39}$$

单选题 10分

设置

At year of ( ), Einstein published his photoelectric effect:

A 1900

B 1905

C 1917

D 1923-1924

提交